



US 20250276761A1

(19) **United States**(12) **Patent Application Publication**
Comer et al.(10) **Pub. No.: US 2025/0276761 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **MICRO-KEEL COOLER FOR MARITIME VEHICLE**(52) **U.S. Cl.**CPC *B63B 3/38* (2013.01); *B63H 21/383* (2013.01); *B63J 2/00* (2013.01)(71) Applicant: **SARONIC TECHNOLOGIES**, Austin, TX (US)(72) Inventors: **Ryan Comer**, Austin, TX (US); **Ryan Howard**, Austin, TX (US); **Trish Anne Roque**, Austin, TX (US); **Robert Duran**, Austin, TX (US); **Douglas Lambert**, Austin, TX (US)

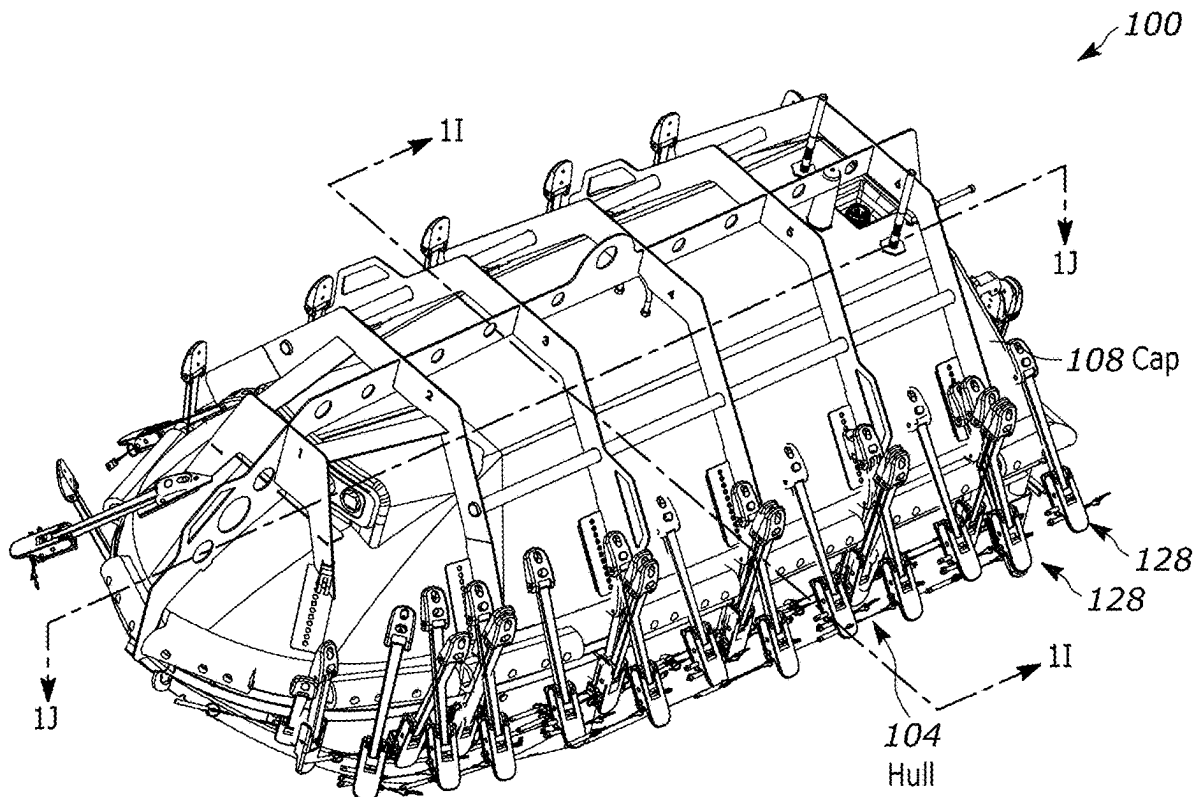
(57)

ABSTRACT

A micro-keel cooler for cooling a coolant flowing through a heat-generating component of a maritime vehicle. The cooler includes an inlet adapted to receive the coolant at a first temperature and an outlet adapted to output the coolant at a second temperature less than the first temperature. The cooler further includes a plurality of tubes fluidly coupled to an inlet mixing chamber of the inlet and an outlet chamber of the outlet and configured to direct the coolant from the inlet mixing chamber to the outlet mixing chamber, and a plate coupled to the inlet and the outlet and disposed adjacent to the tubes, the plate configured to direct a fluid over at least a portion of the tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a cooling temperature that is less than the first temperature.

(21) Appl. No.: **19/070,242**(22) Filed: **Mar. 4, 2025****Related U.S. Application Data**

(60) Provisional application No. 63/561,181, filed on Mar. 4, 2024.

Publication Classification(51) **Int. Cl.***B63B 3/38* (2006.01)*B63H 21/38* (2006.01)*B63J 2/00* (2006.01)

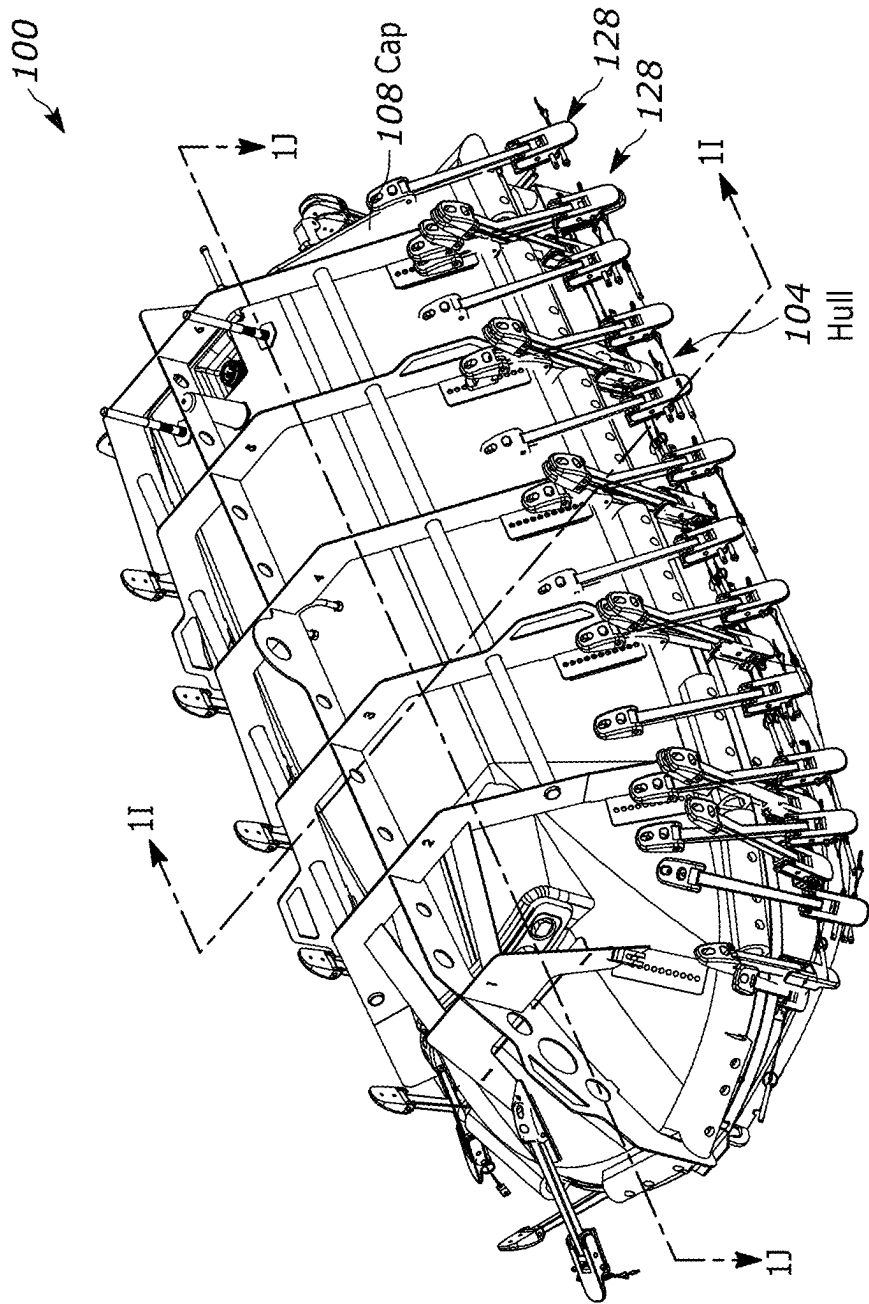


FIG. 1A

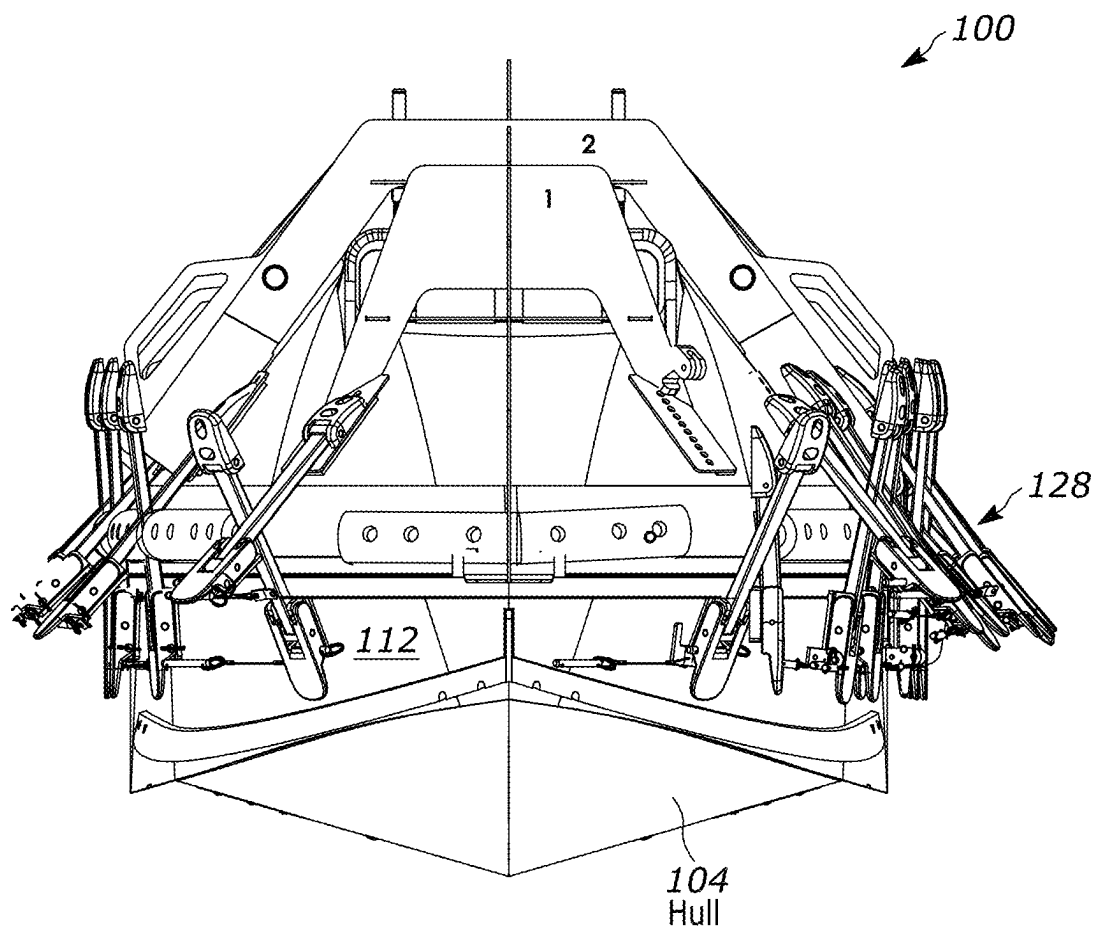


FIG. 1B

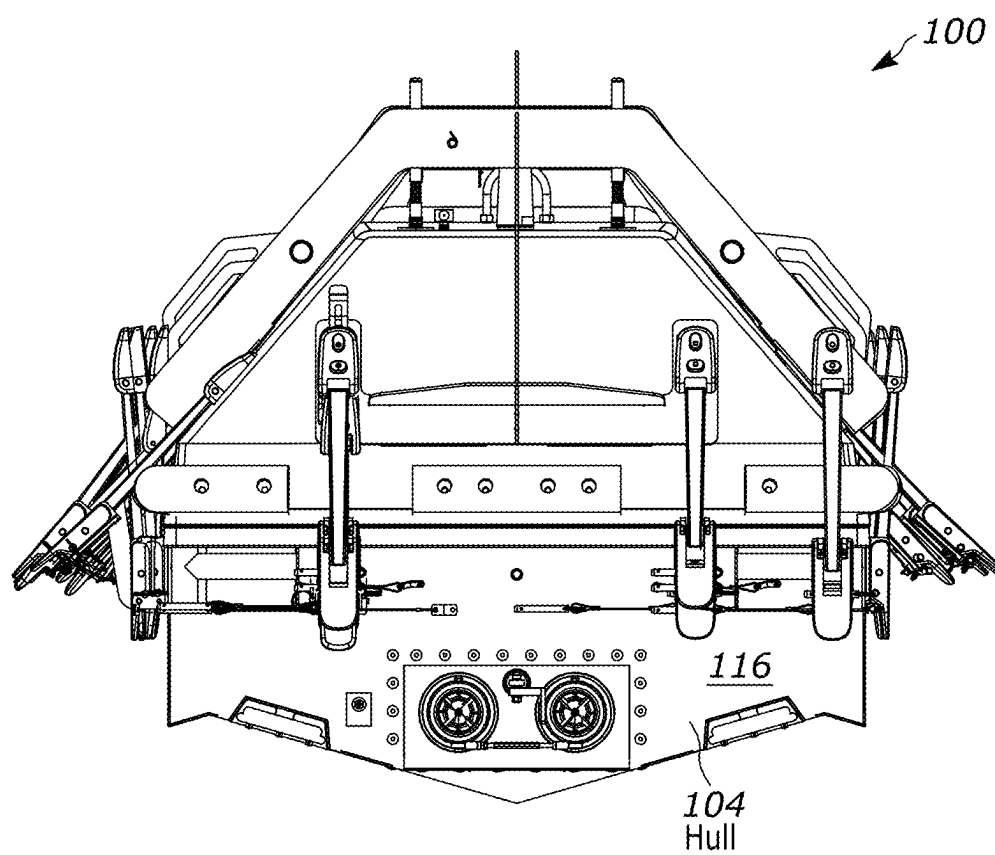


FIG. 1C

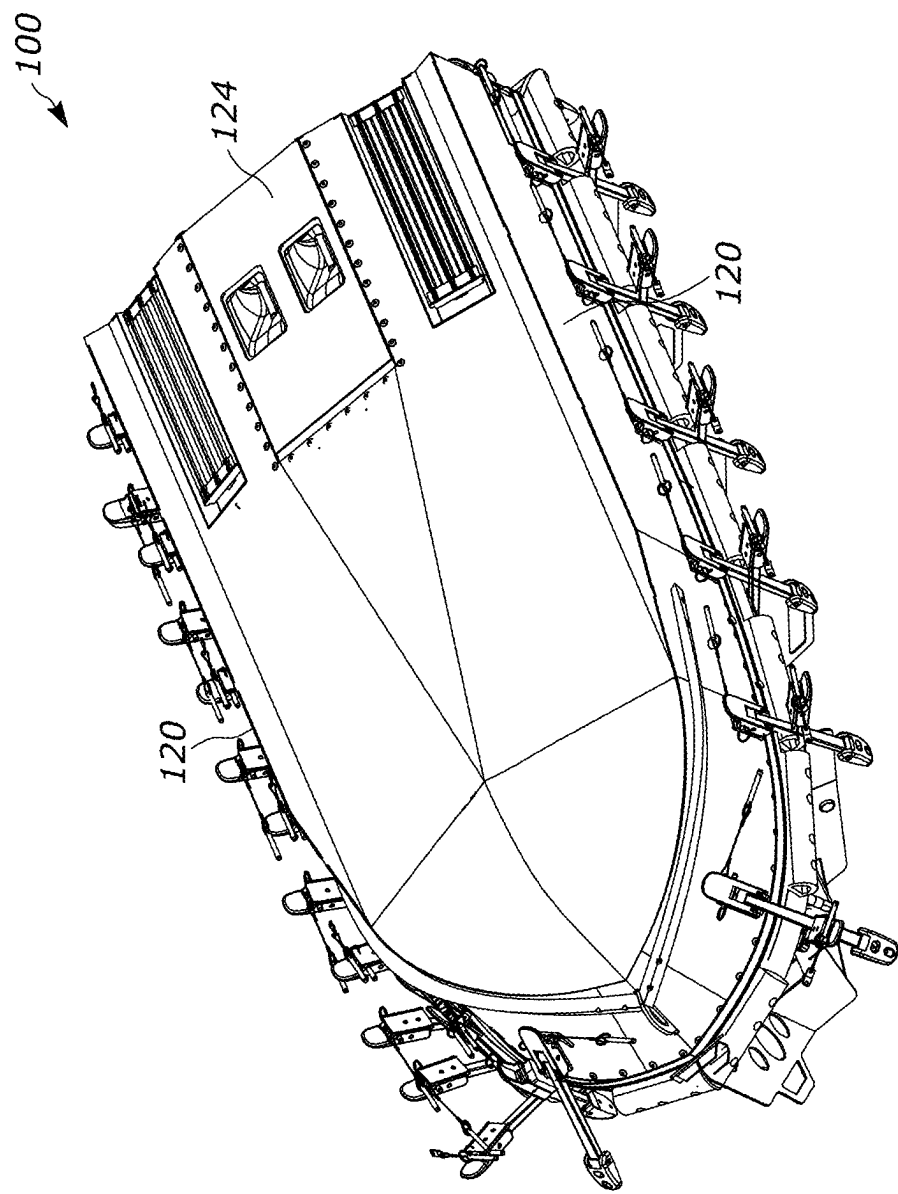


FIG. 1D

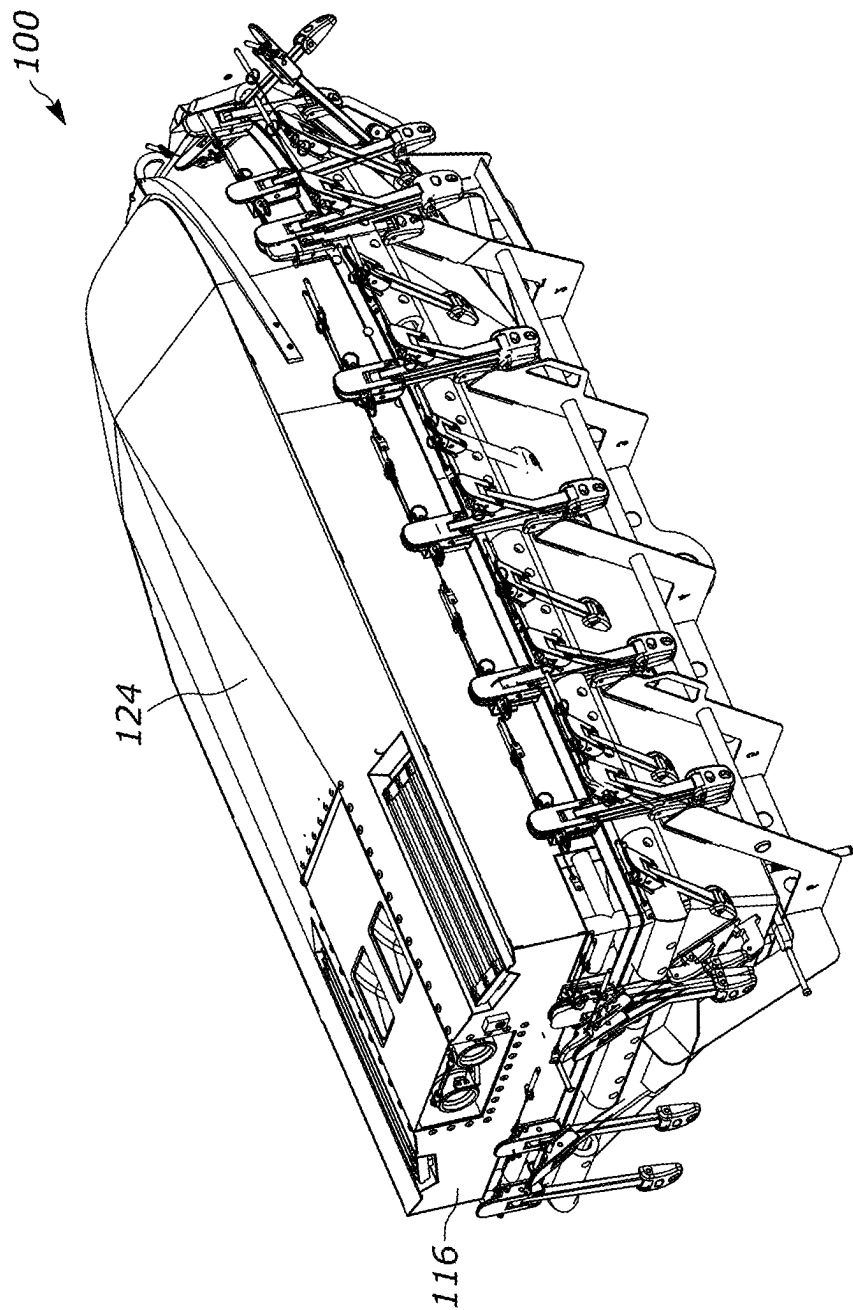


FIG. 1E

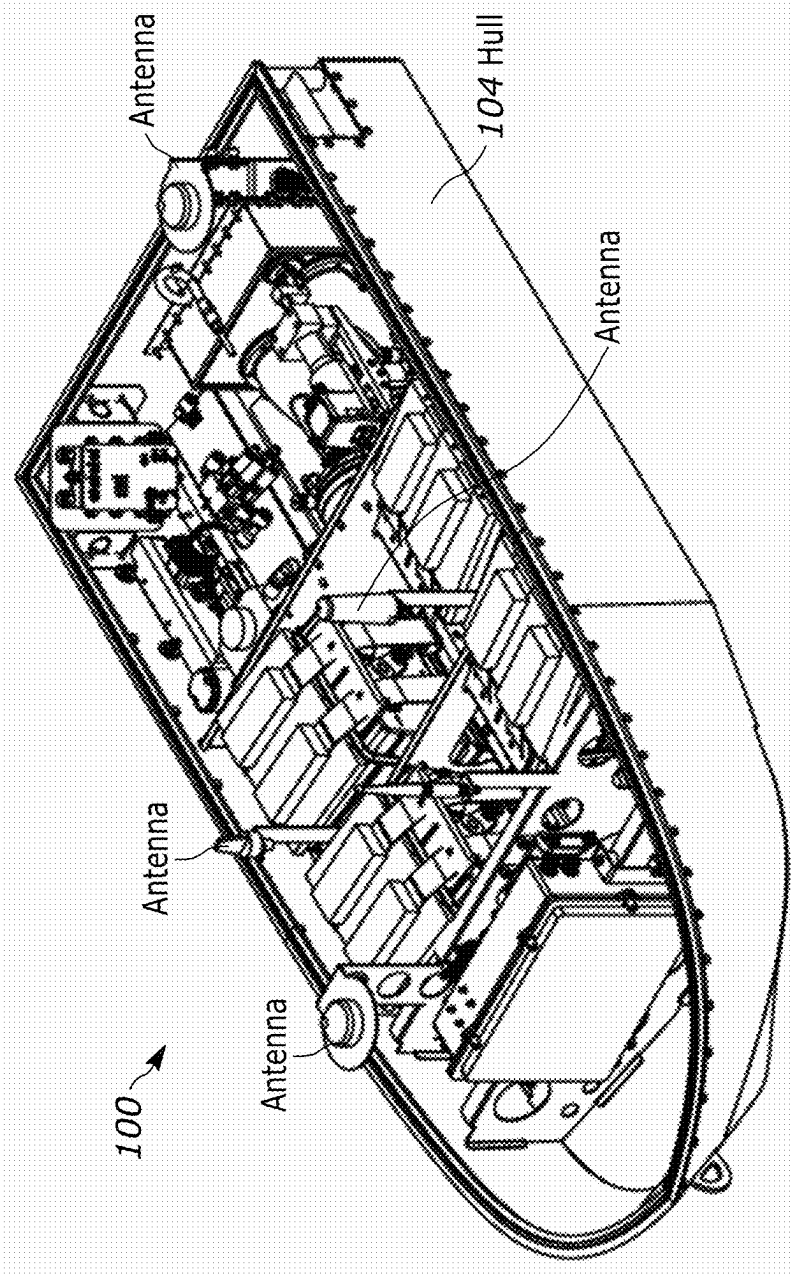


FIG. 1F

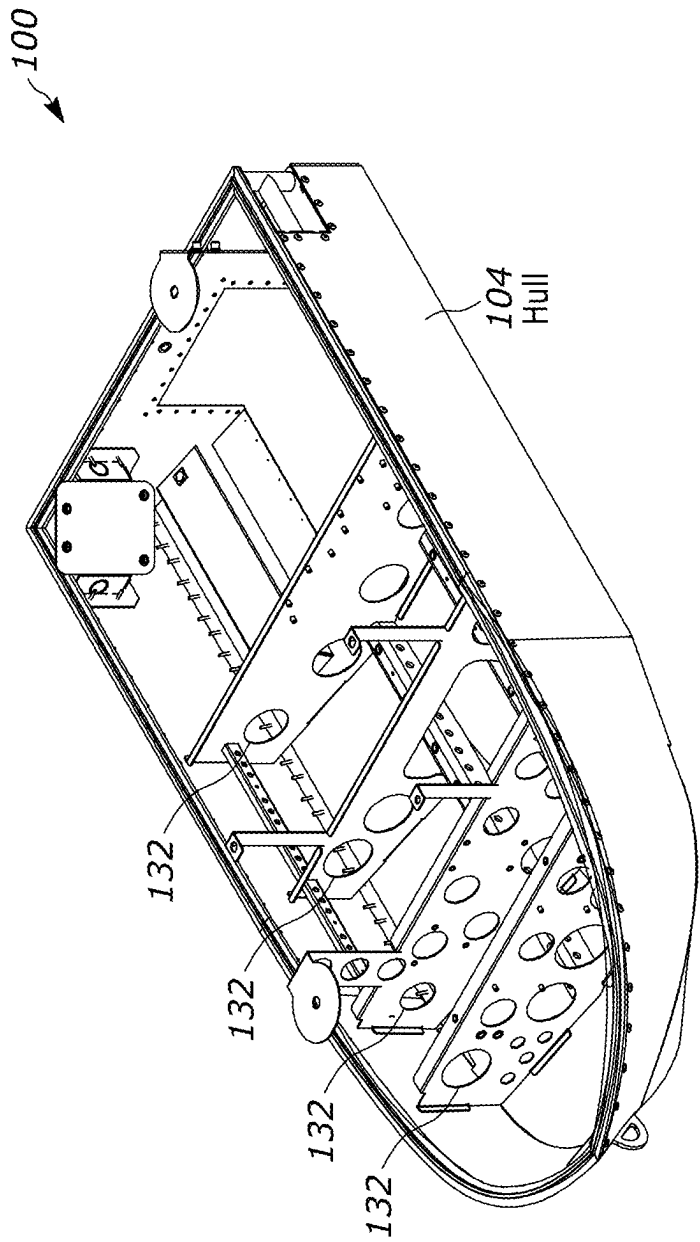


FIG. 1G

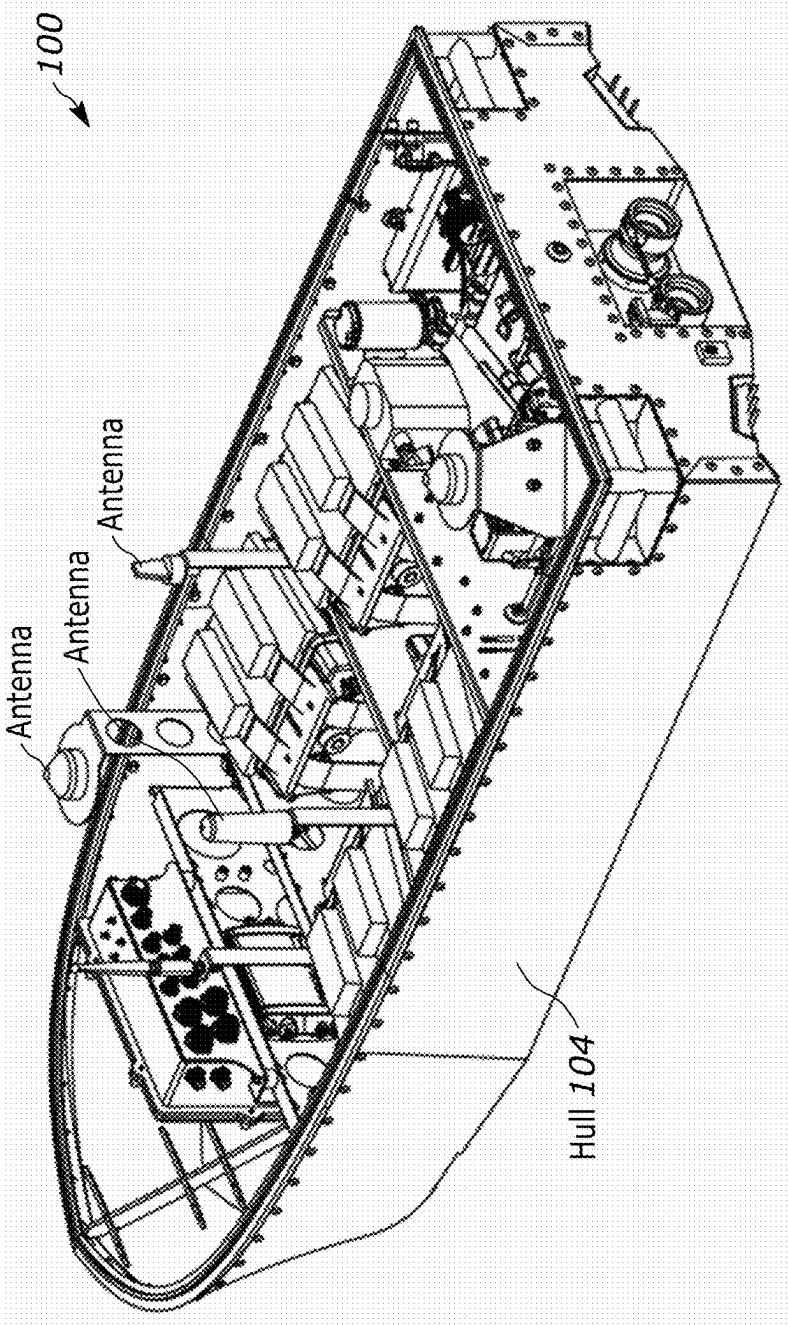


FIG. 1H

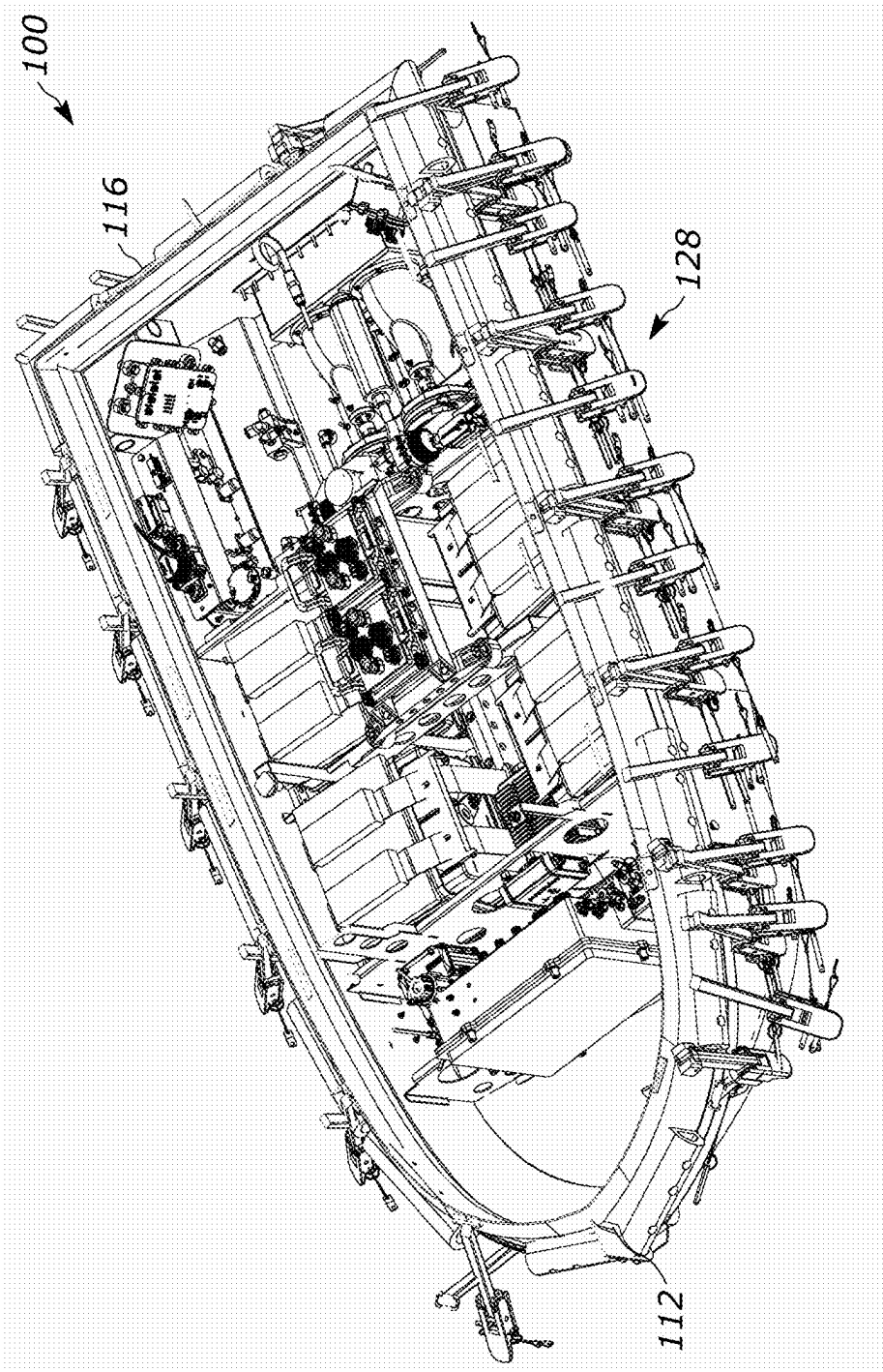


FIG. 11

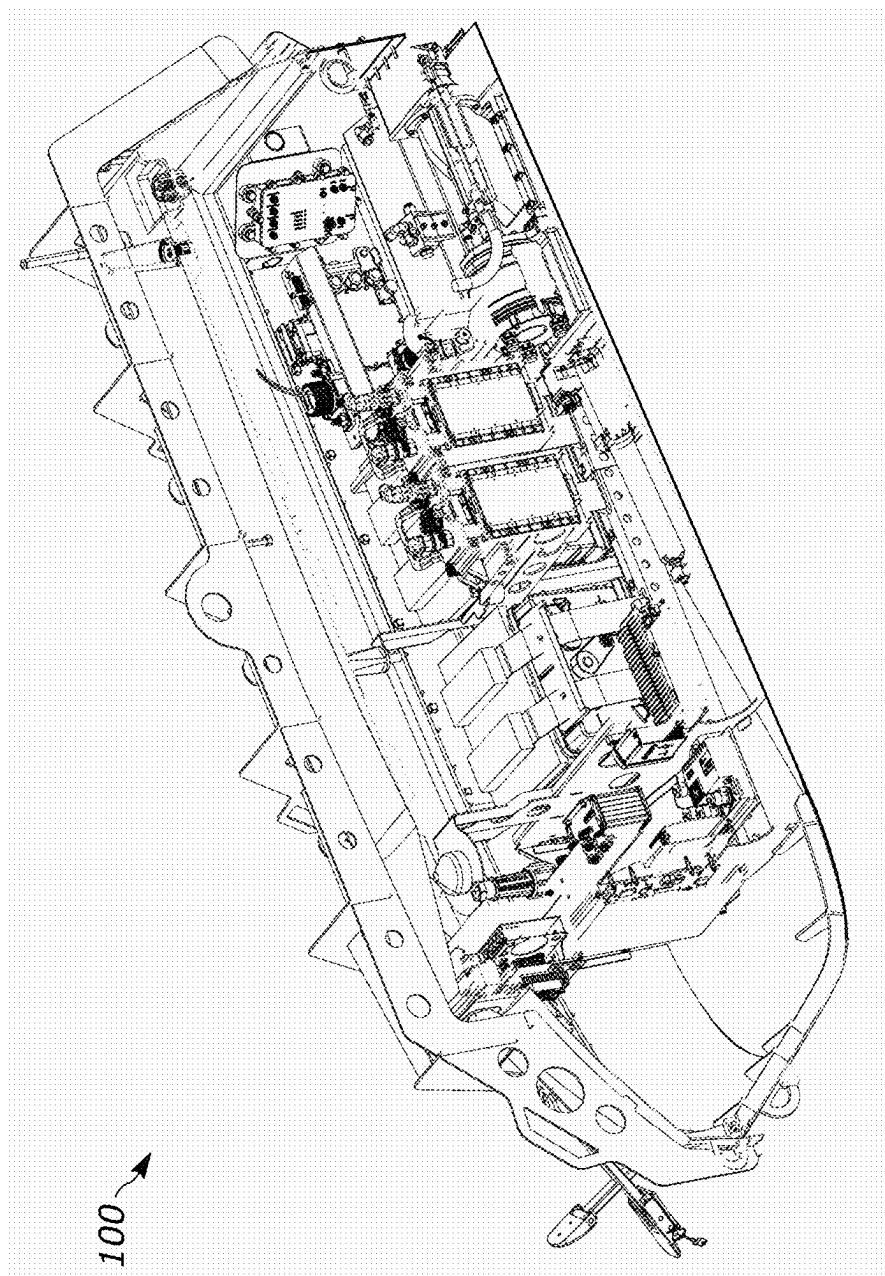


FIG. 1J

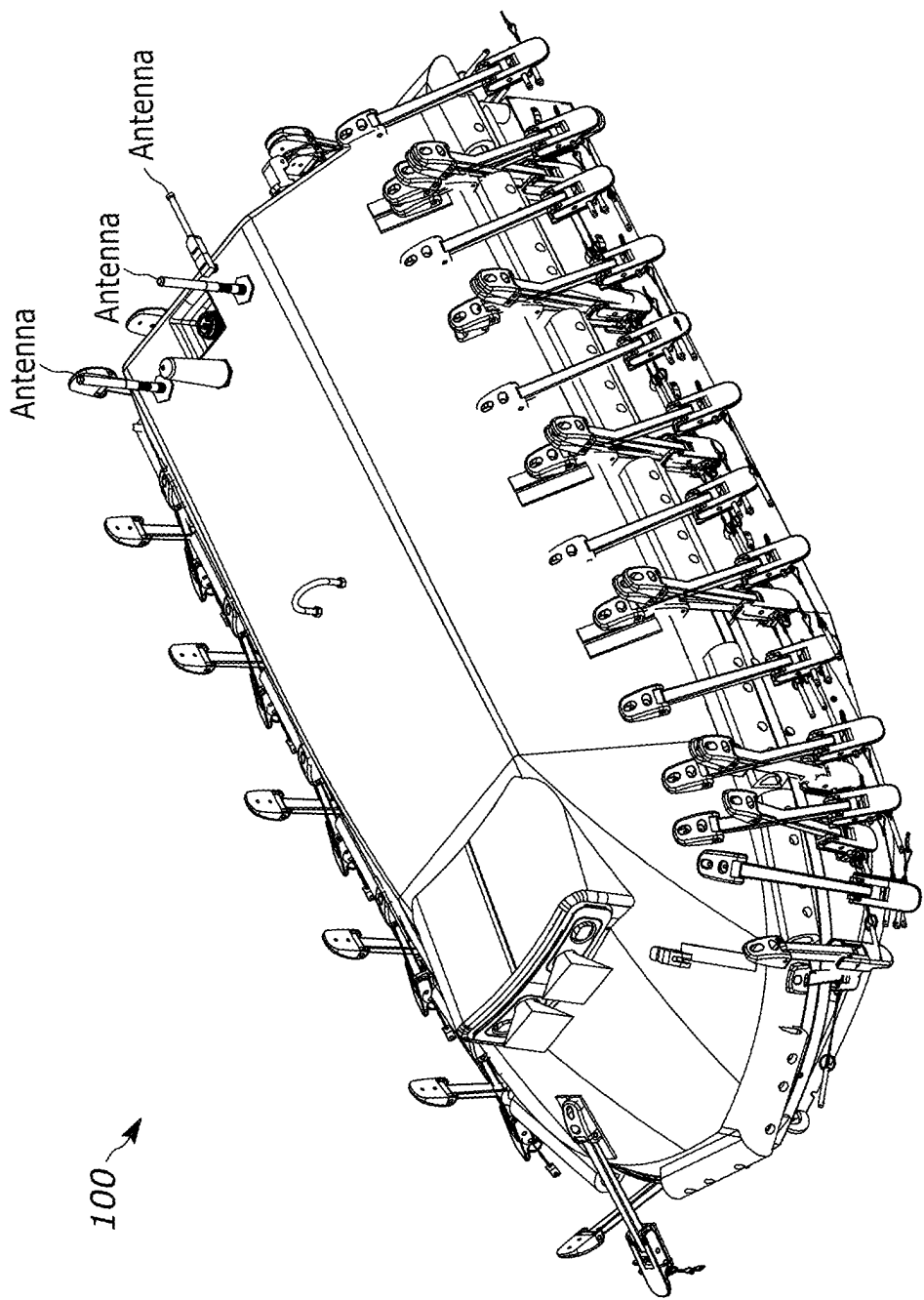


FIG. 1K

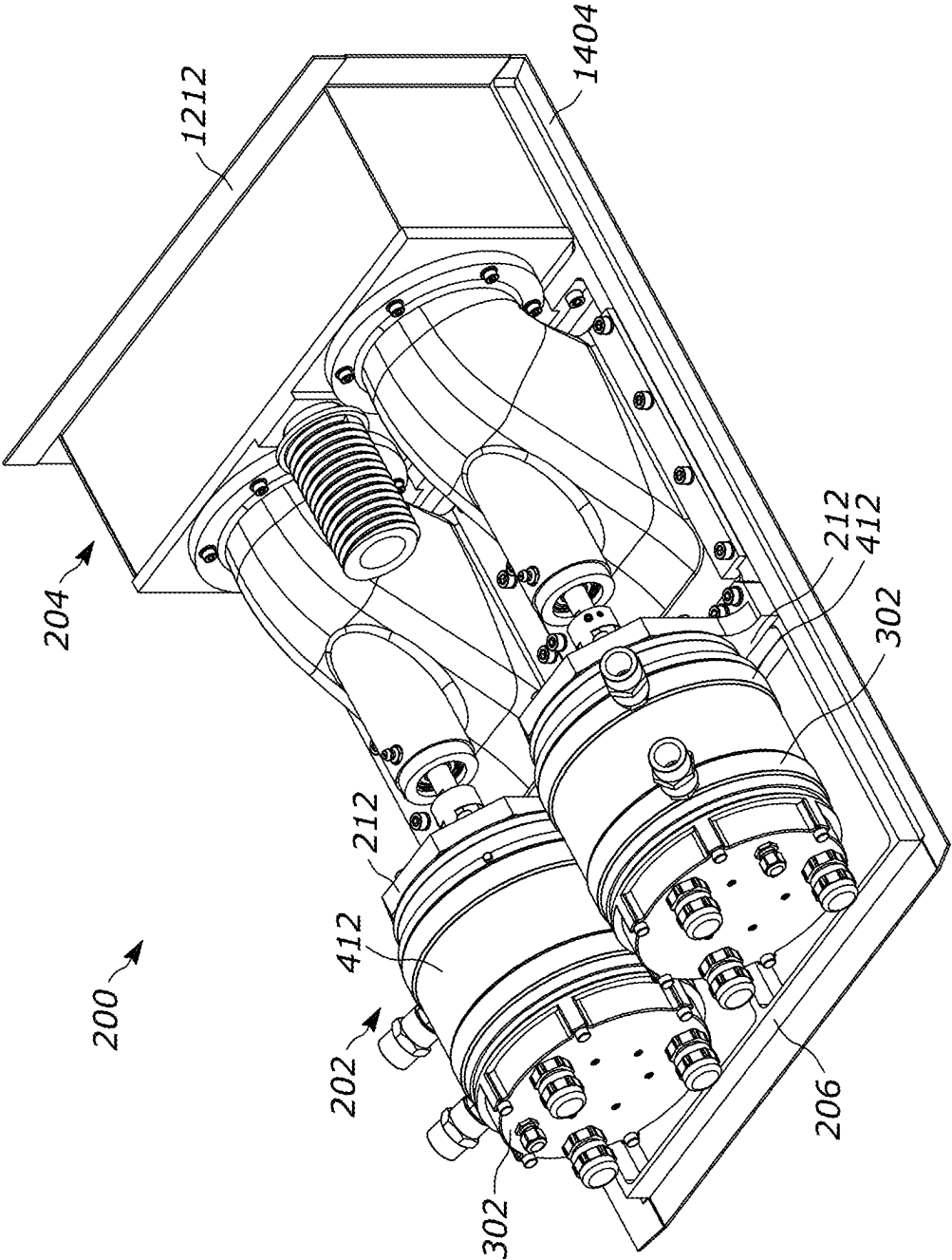


FIG. 2

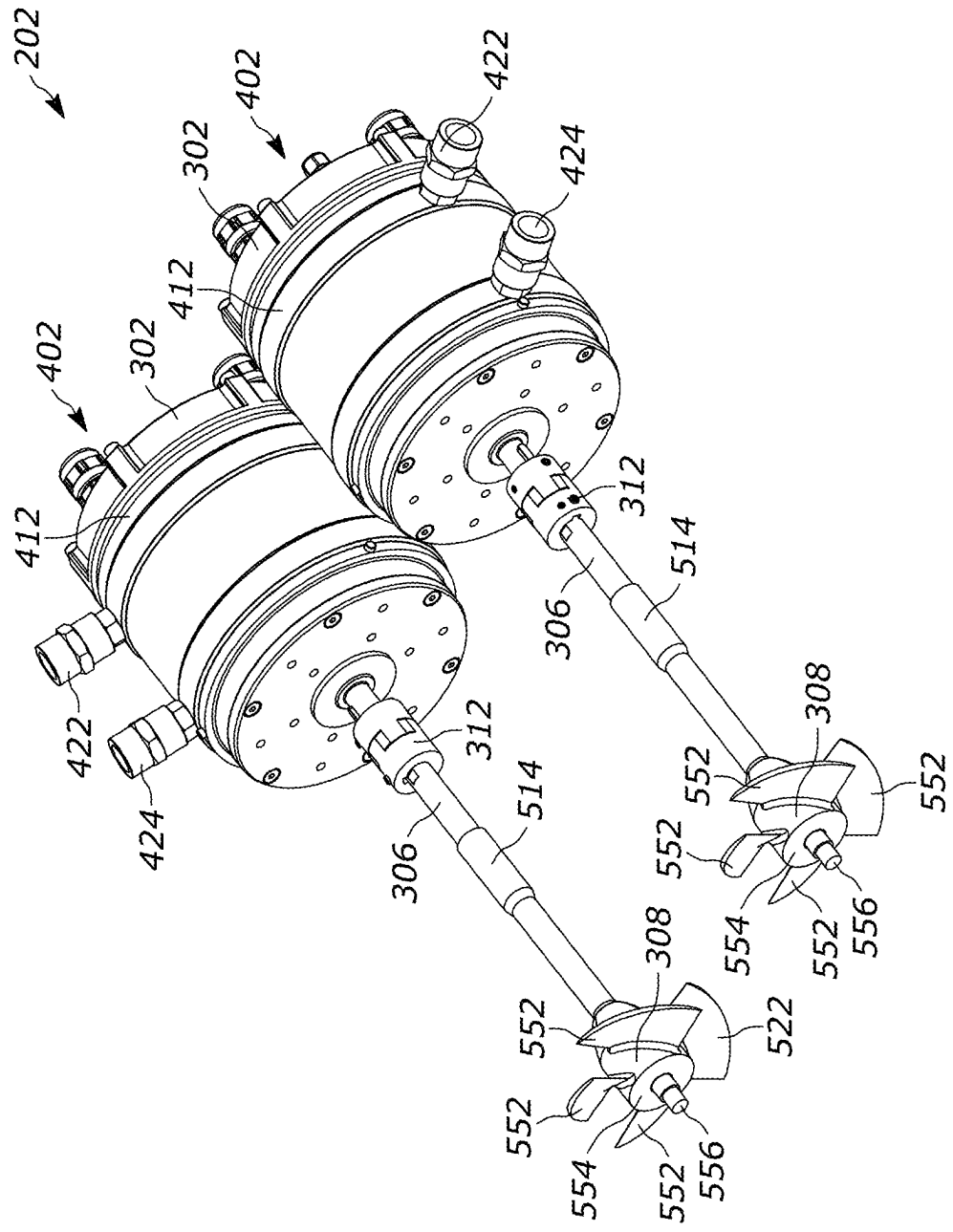


FIG. 3

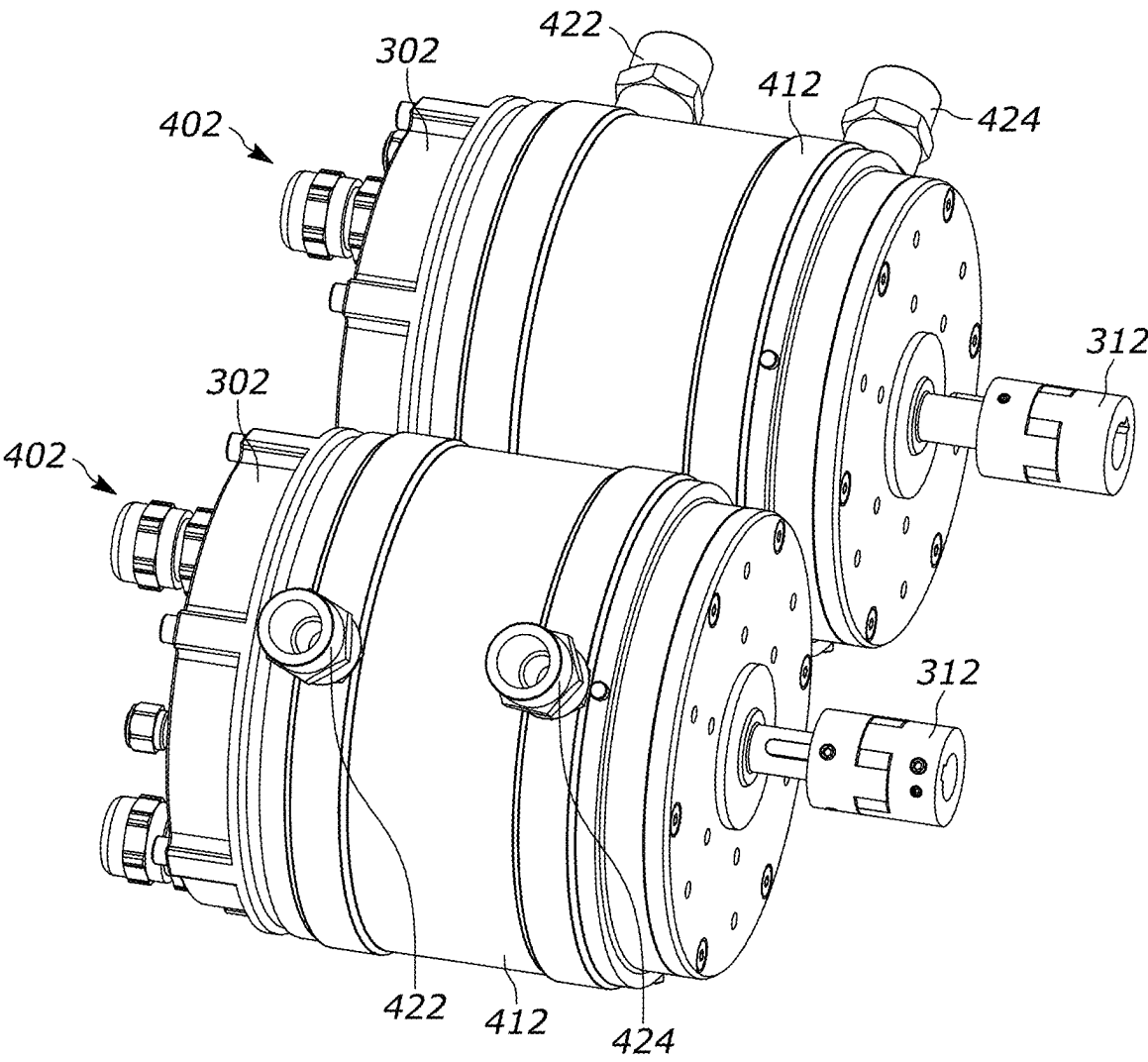


FIG. 4

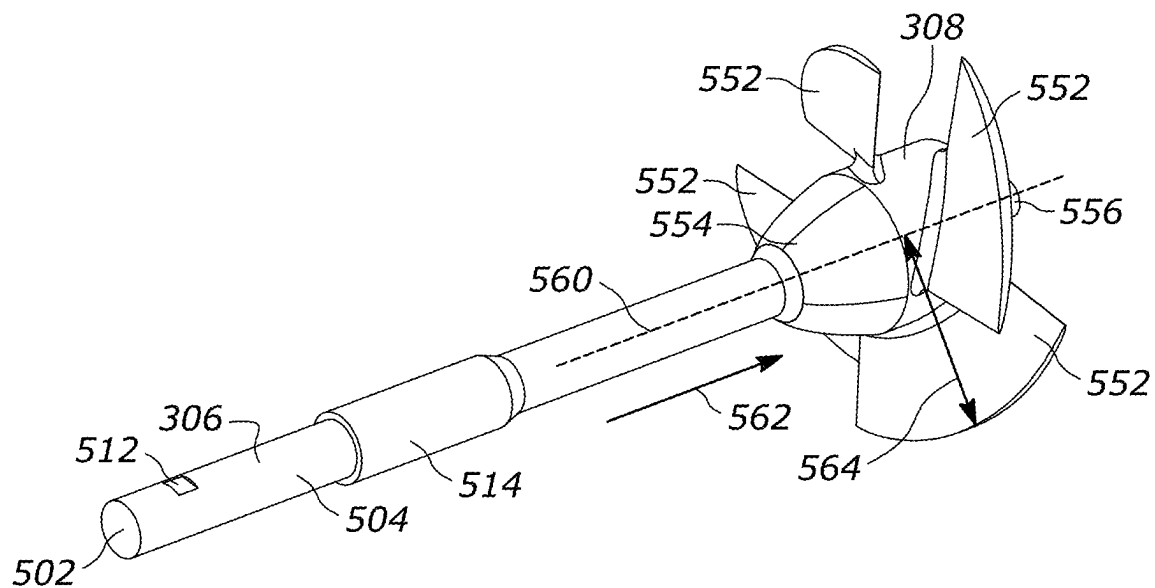


FIG. 5

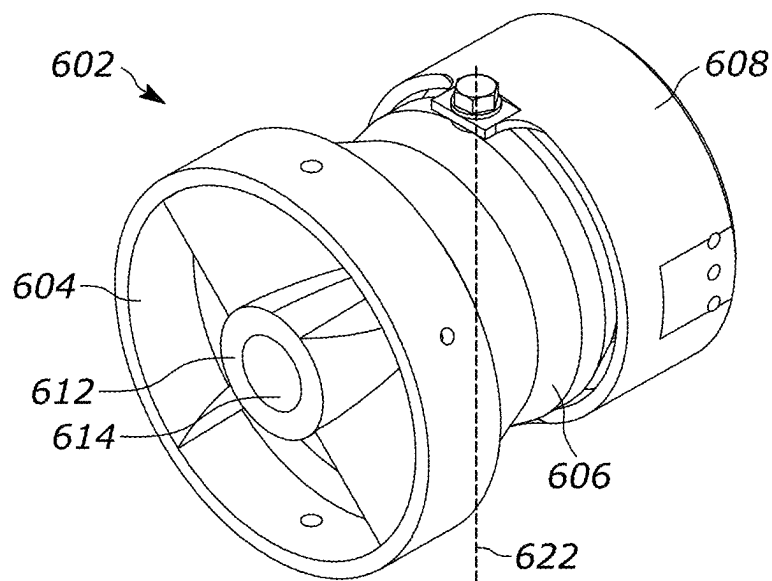


FIG. 6

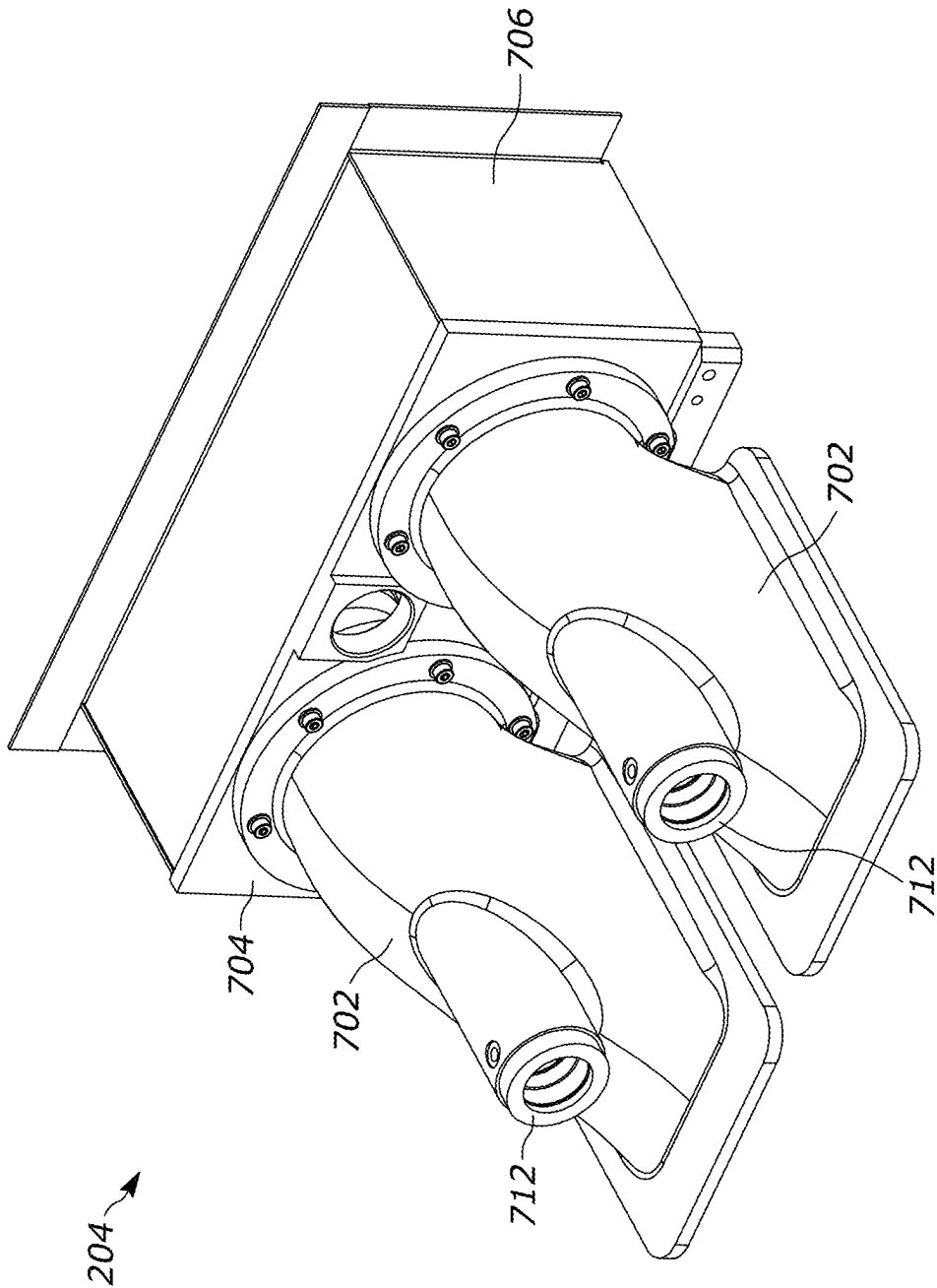


FIG. 7

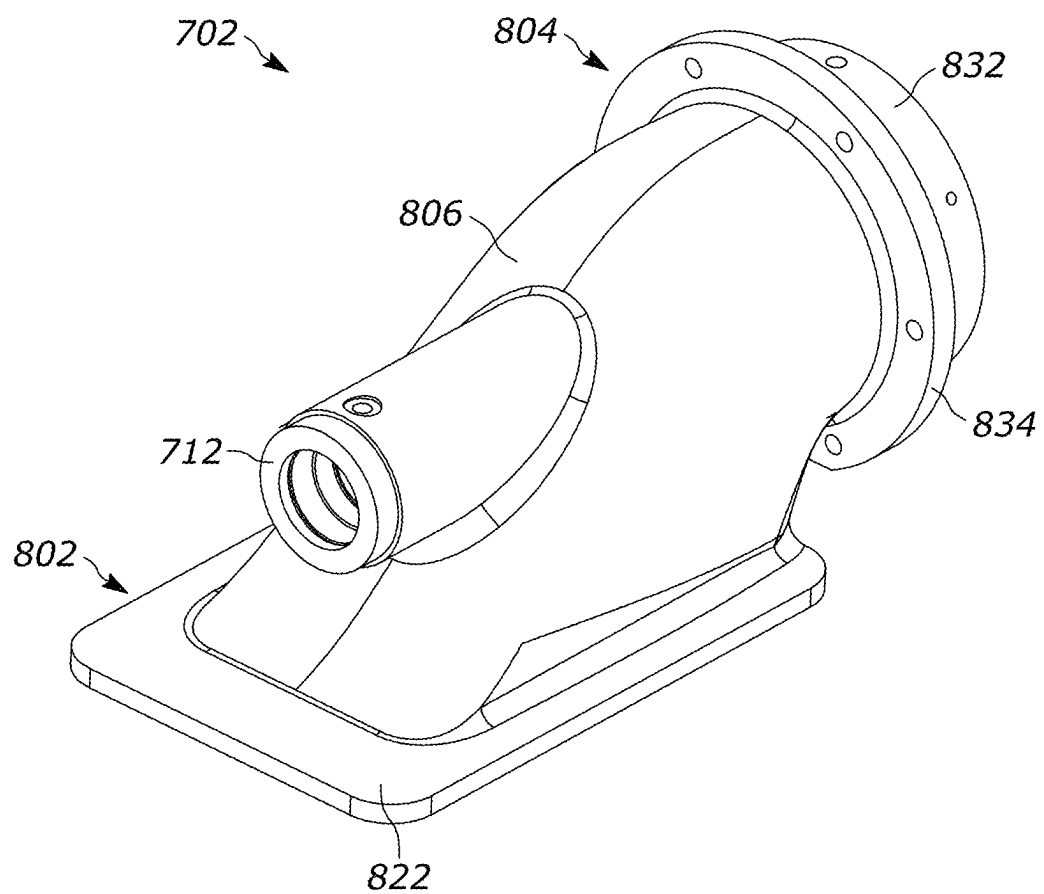


FIG. 8

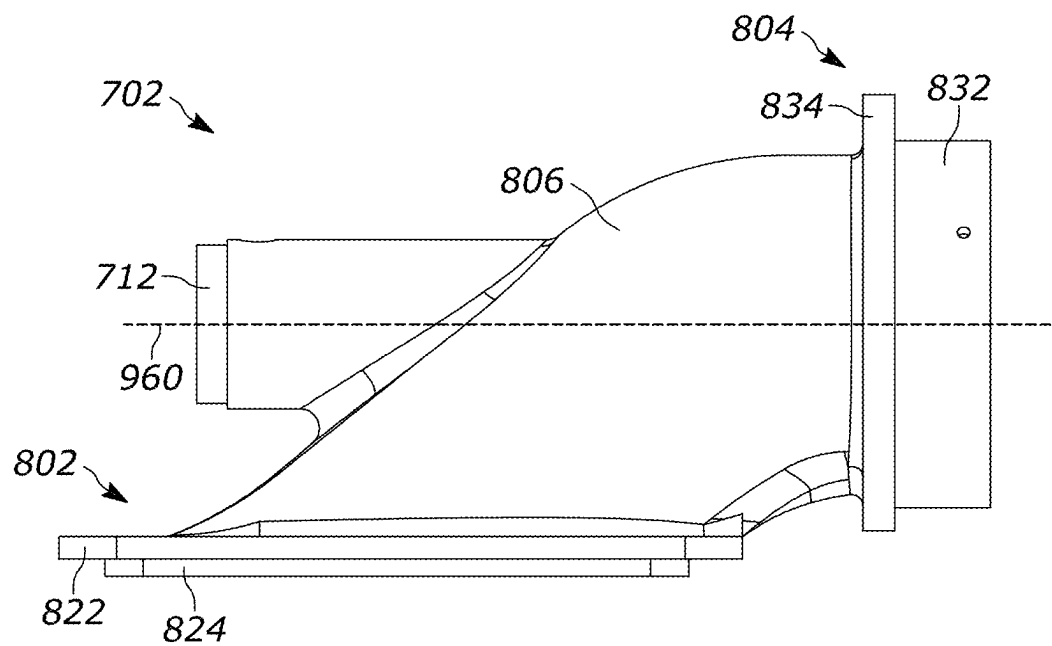


FIG. 9

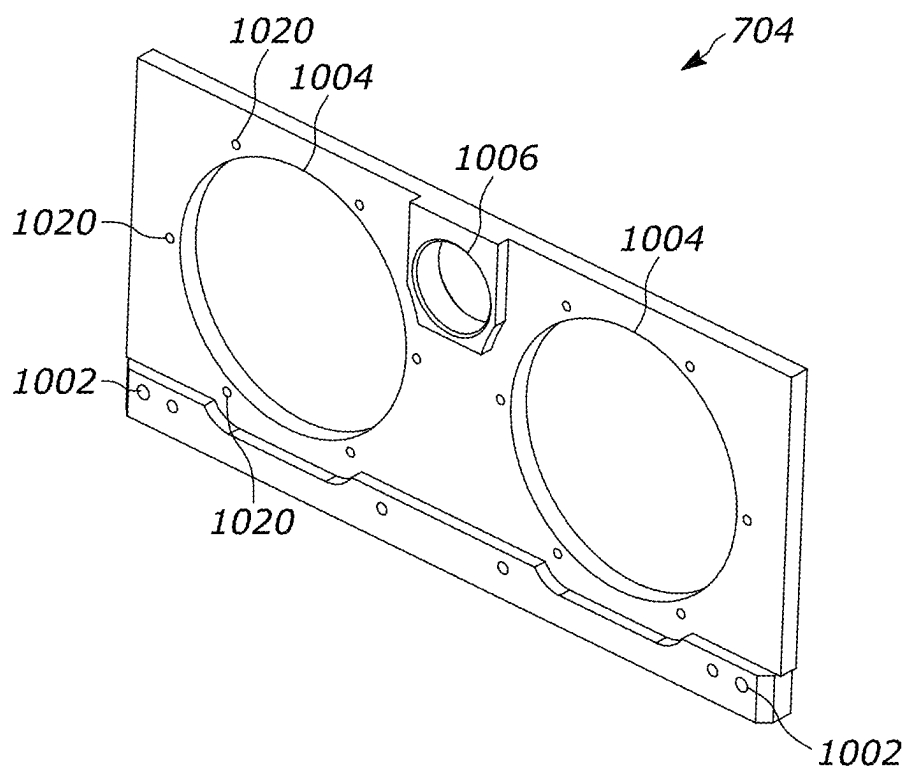


FIG. 10A

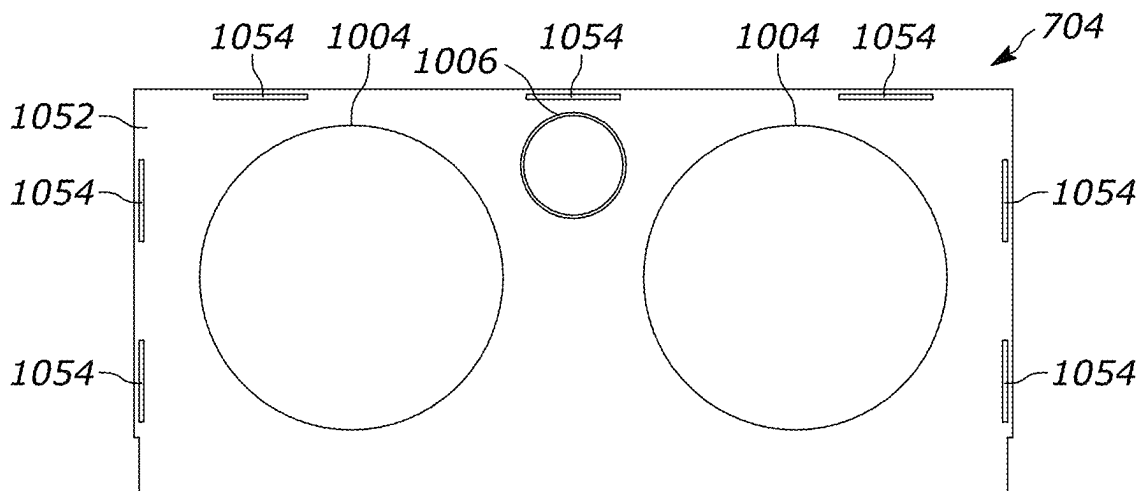


FIG. 10B

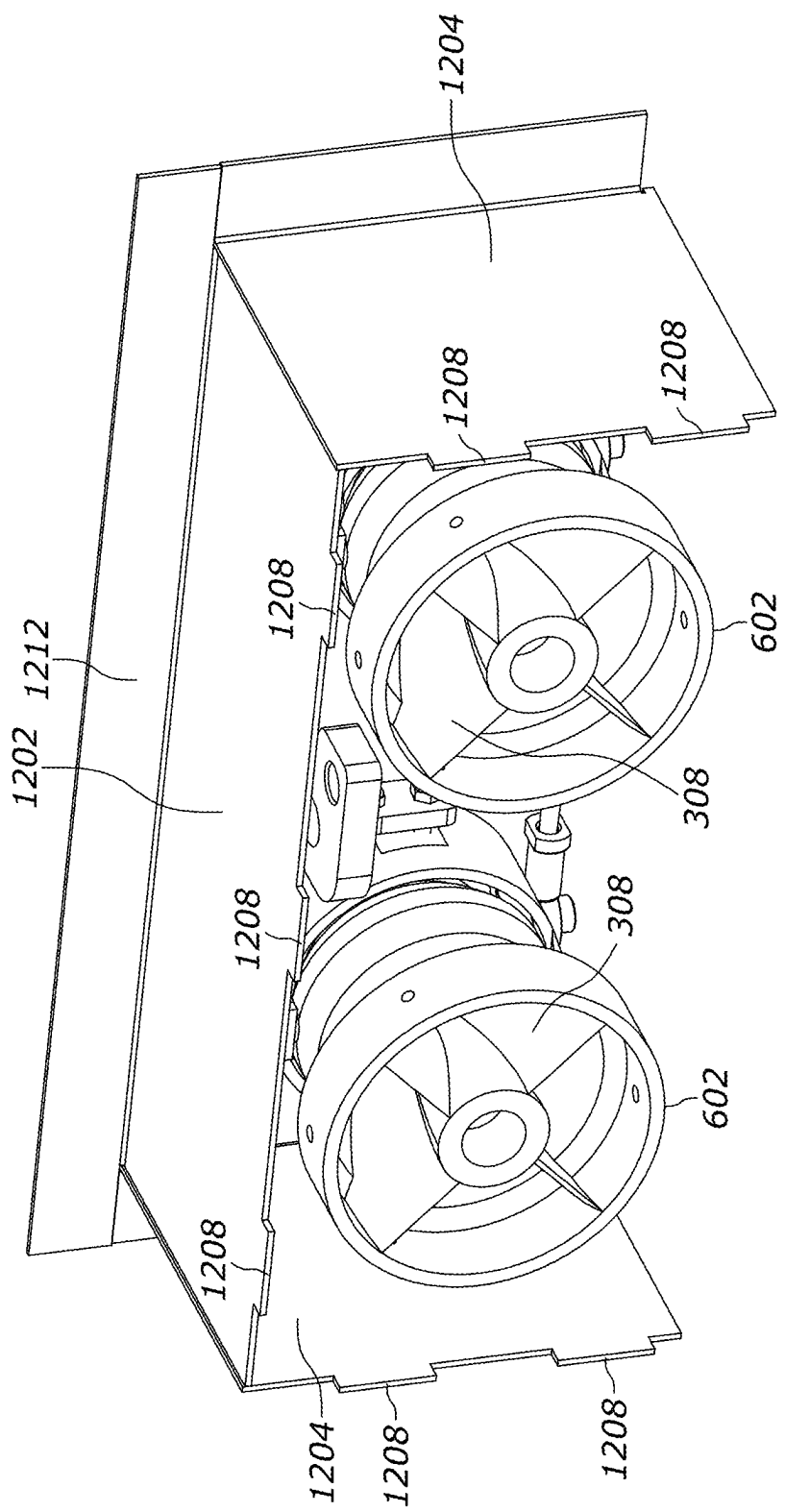


FIG. 11

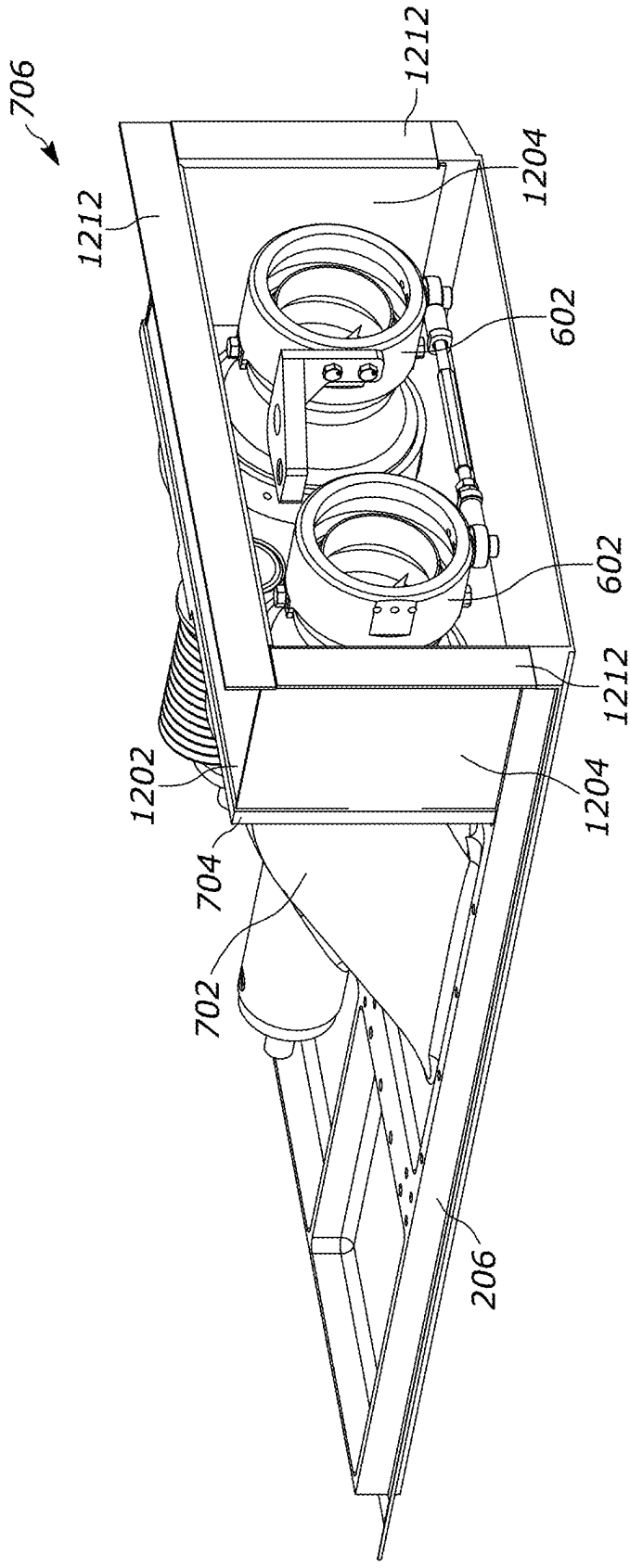


FIG. 12

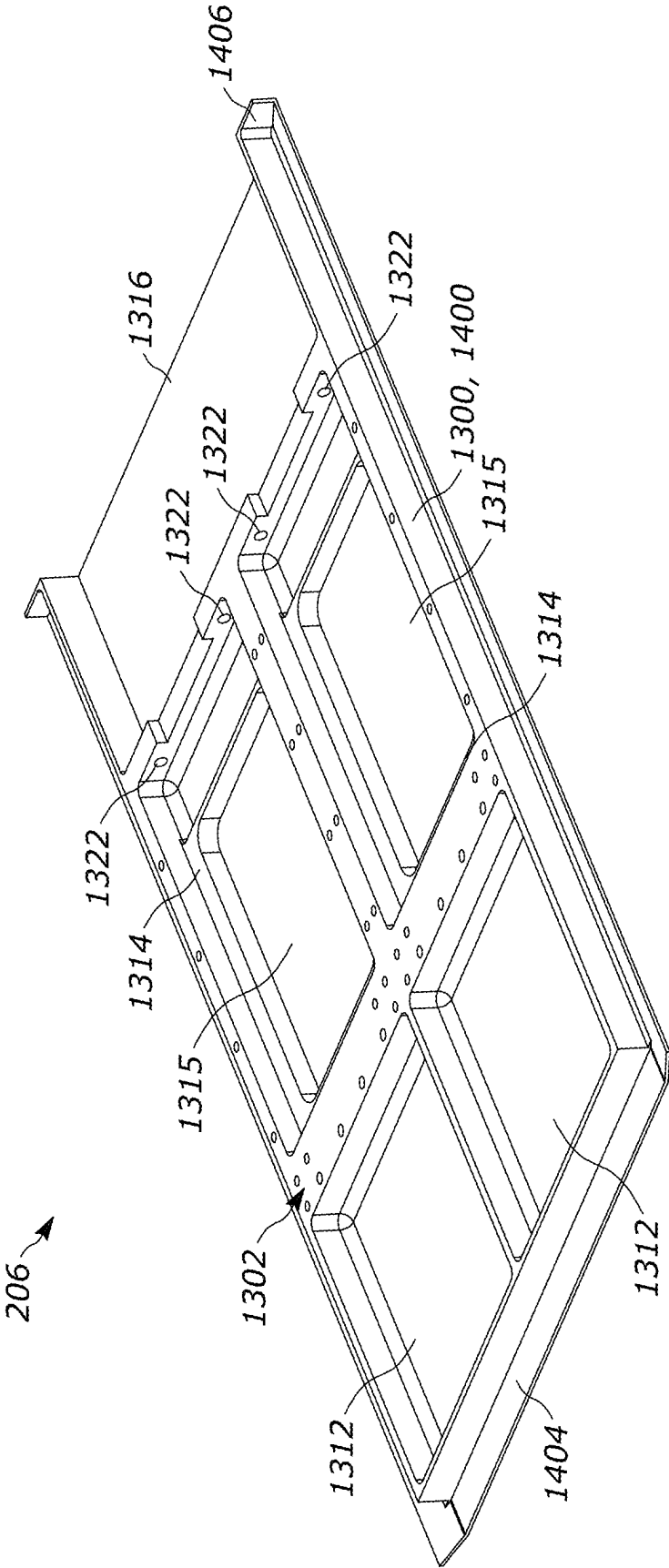


FIG. 13A

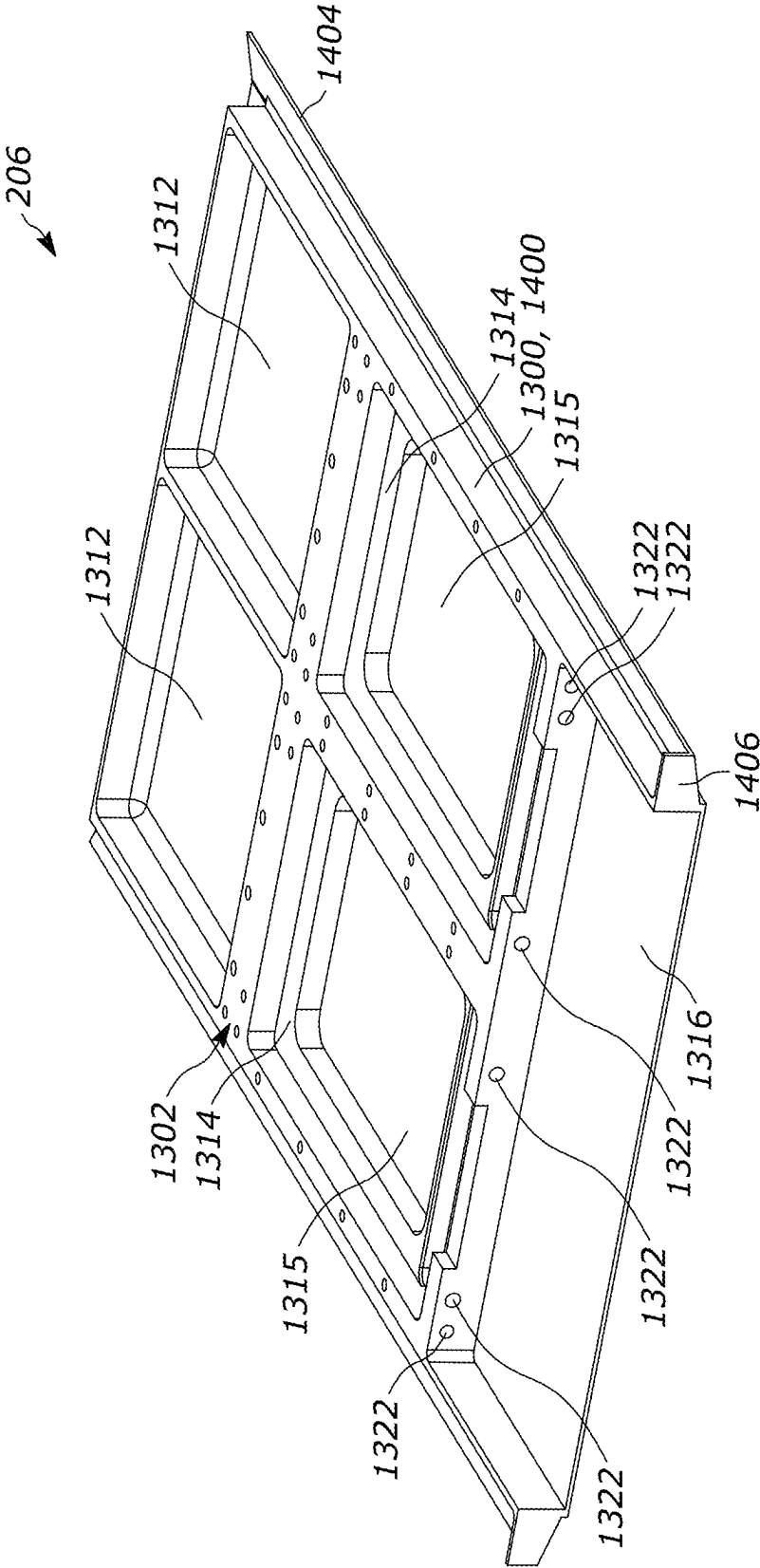


FIG. 13B

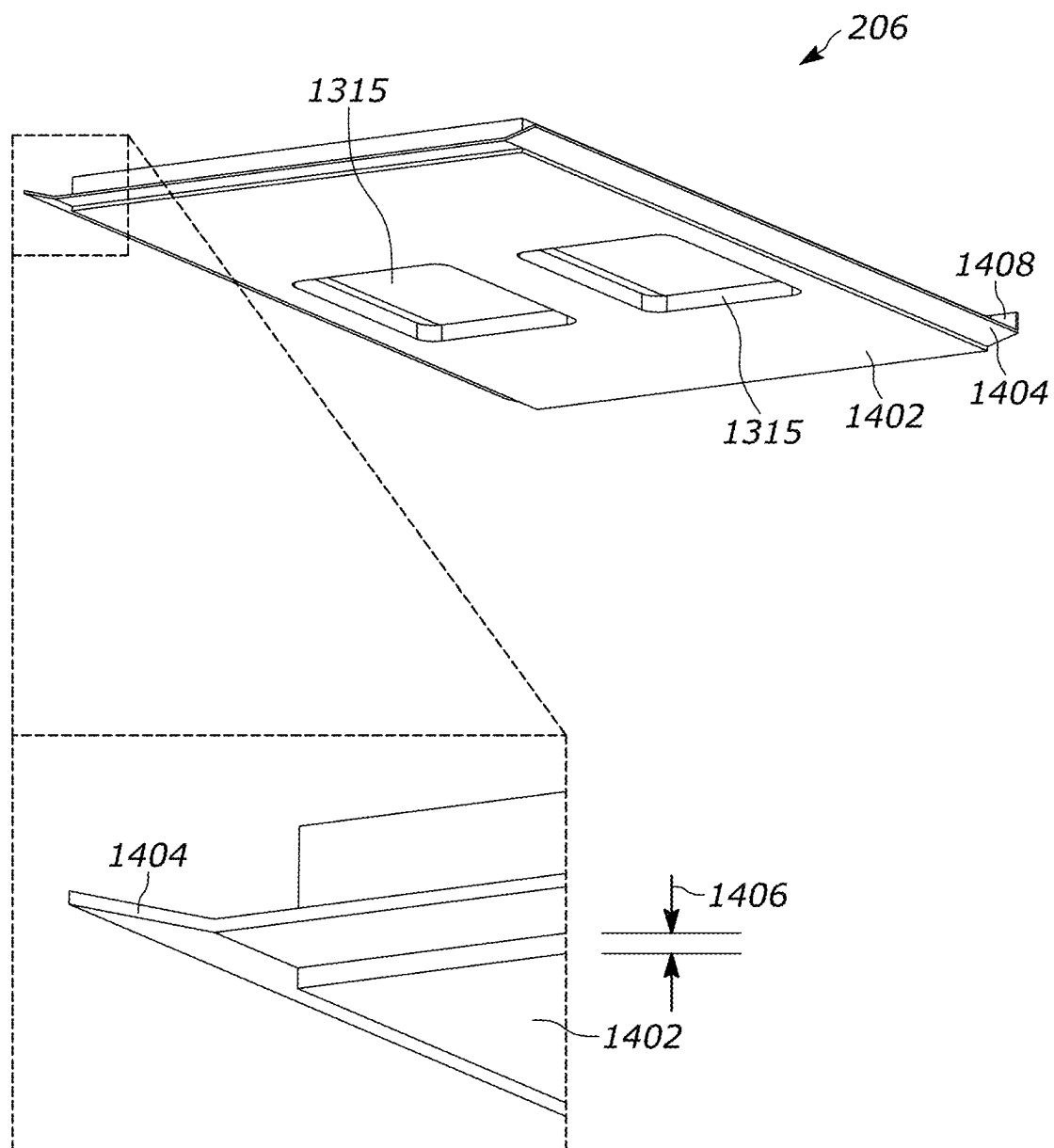


FIG. 14

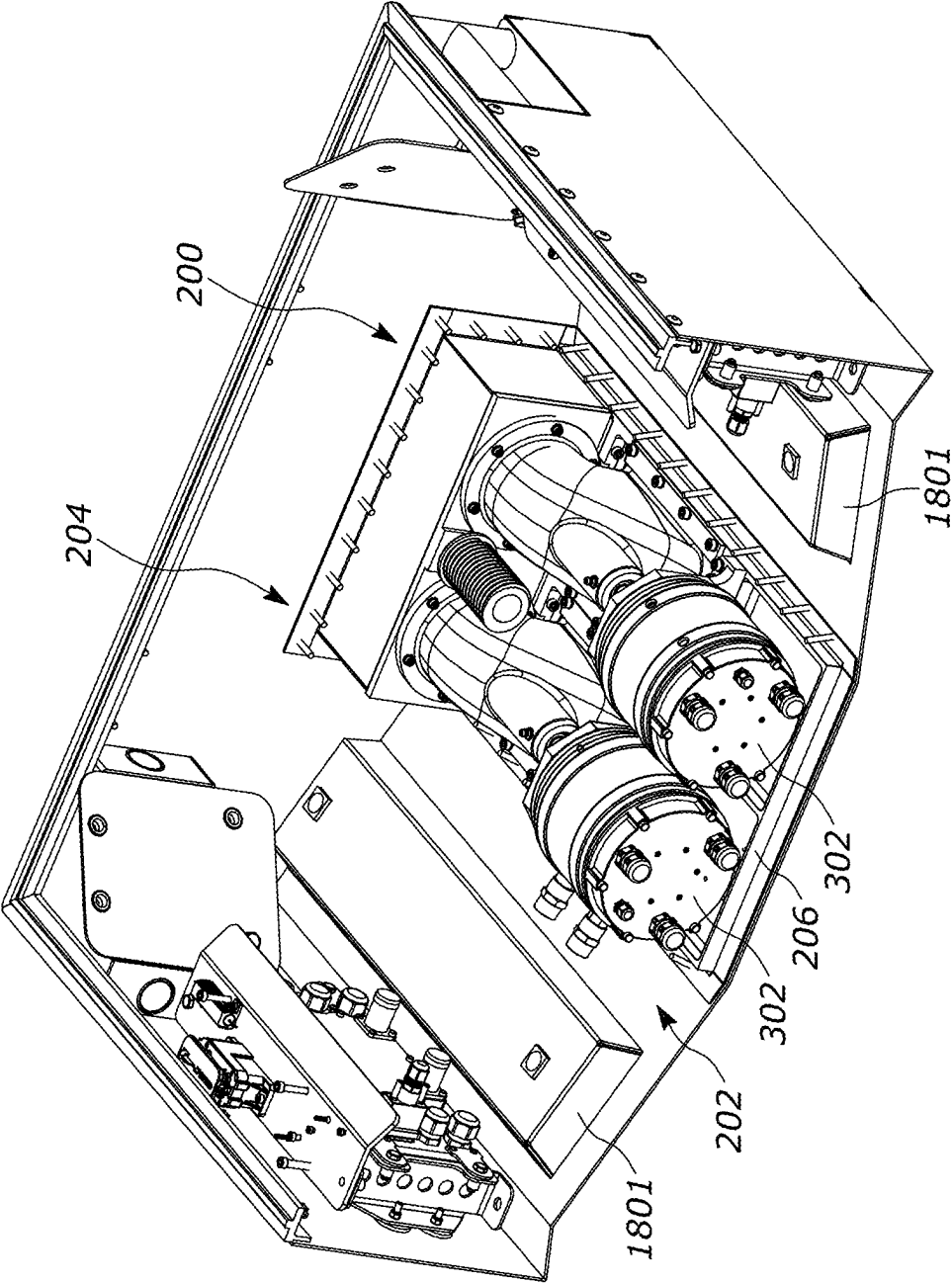


FIG. 15

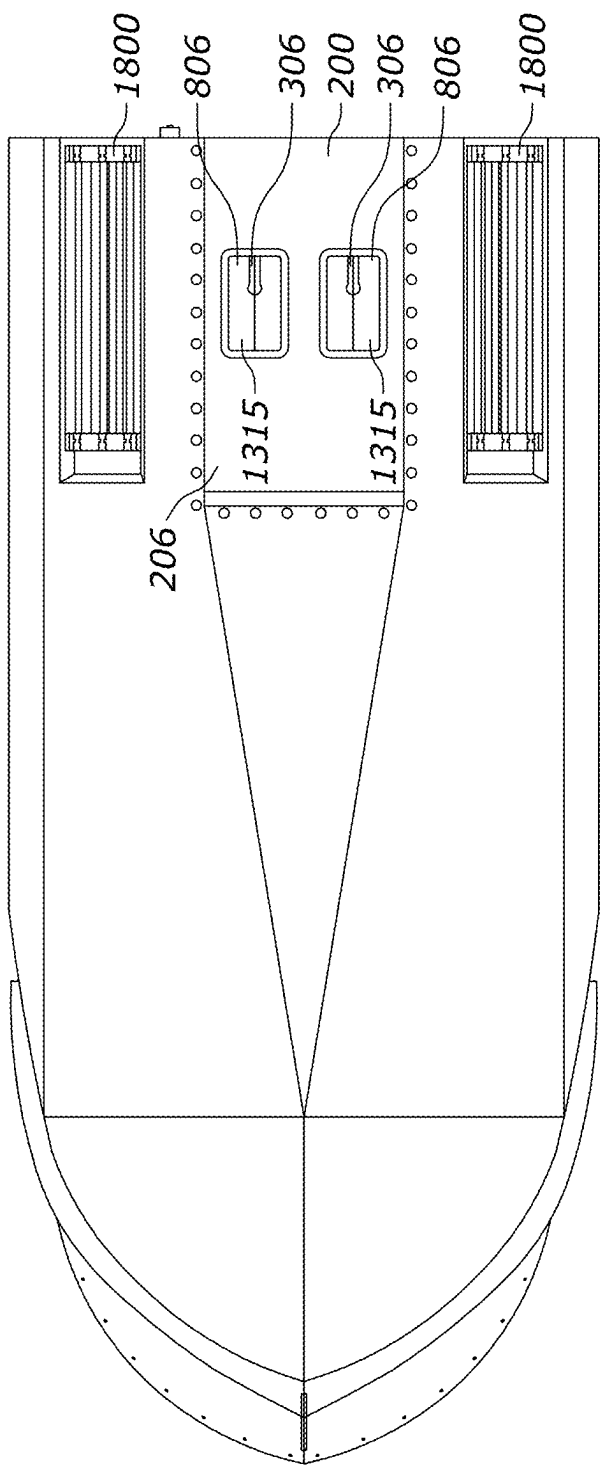


FIG. 16

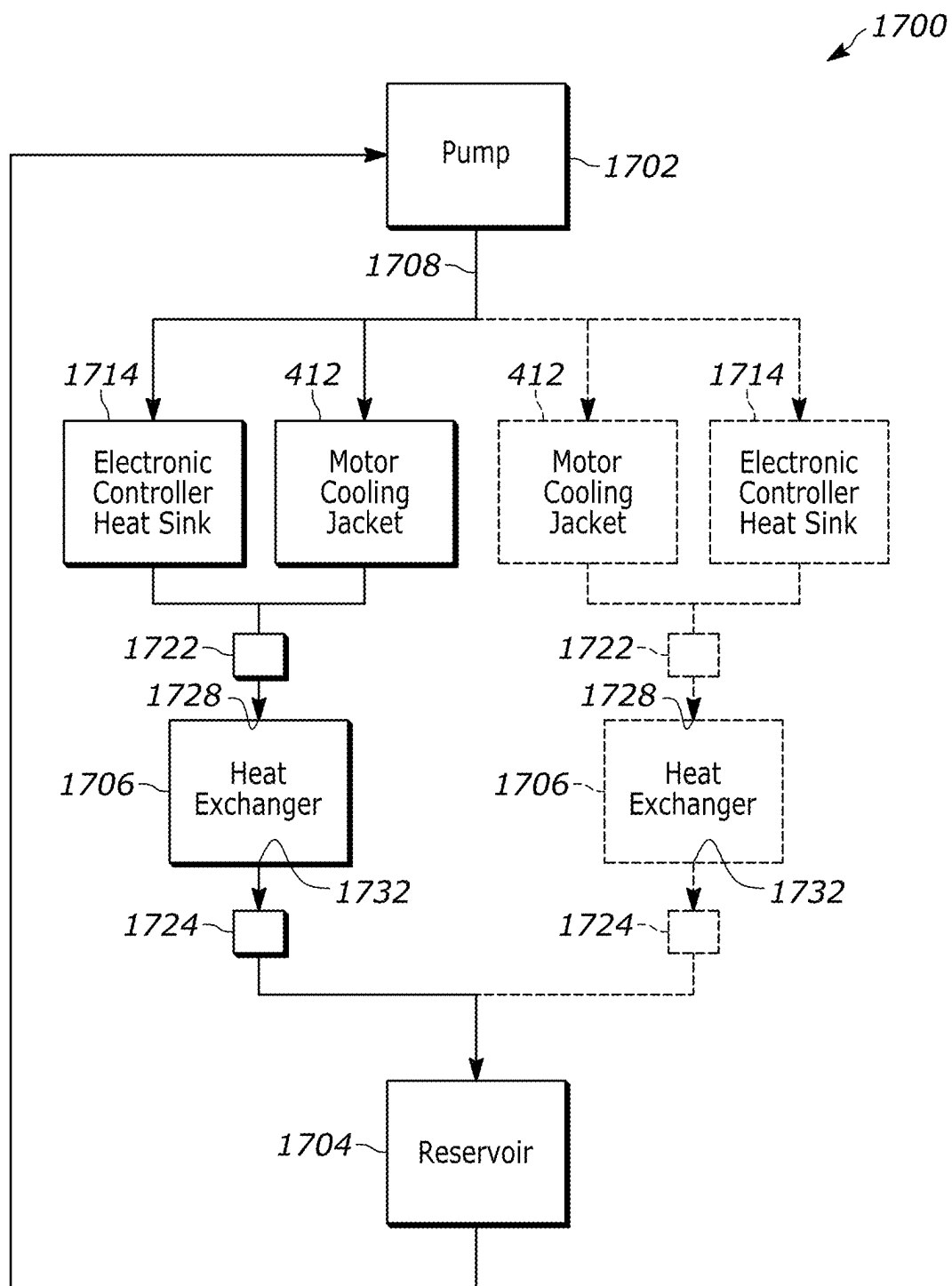


FIG. 17

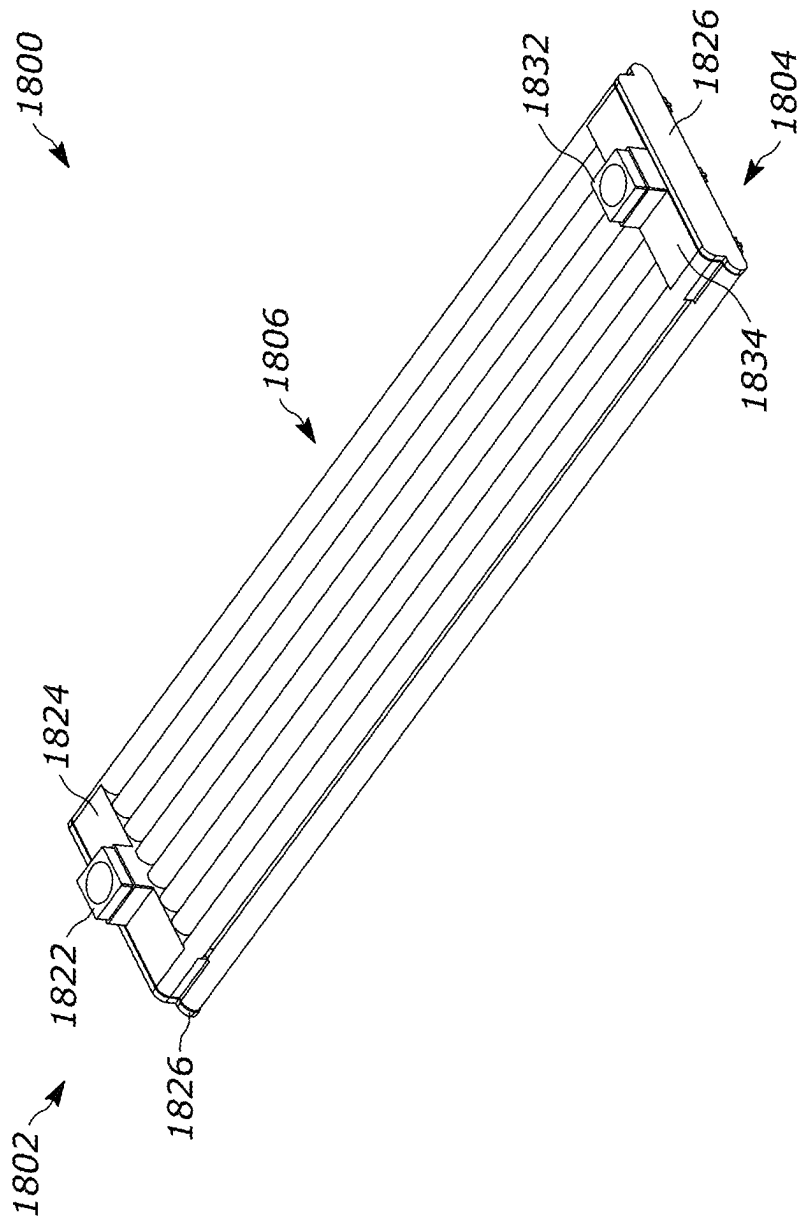


FIG. 18A

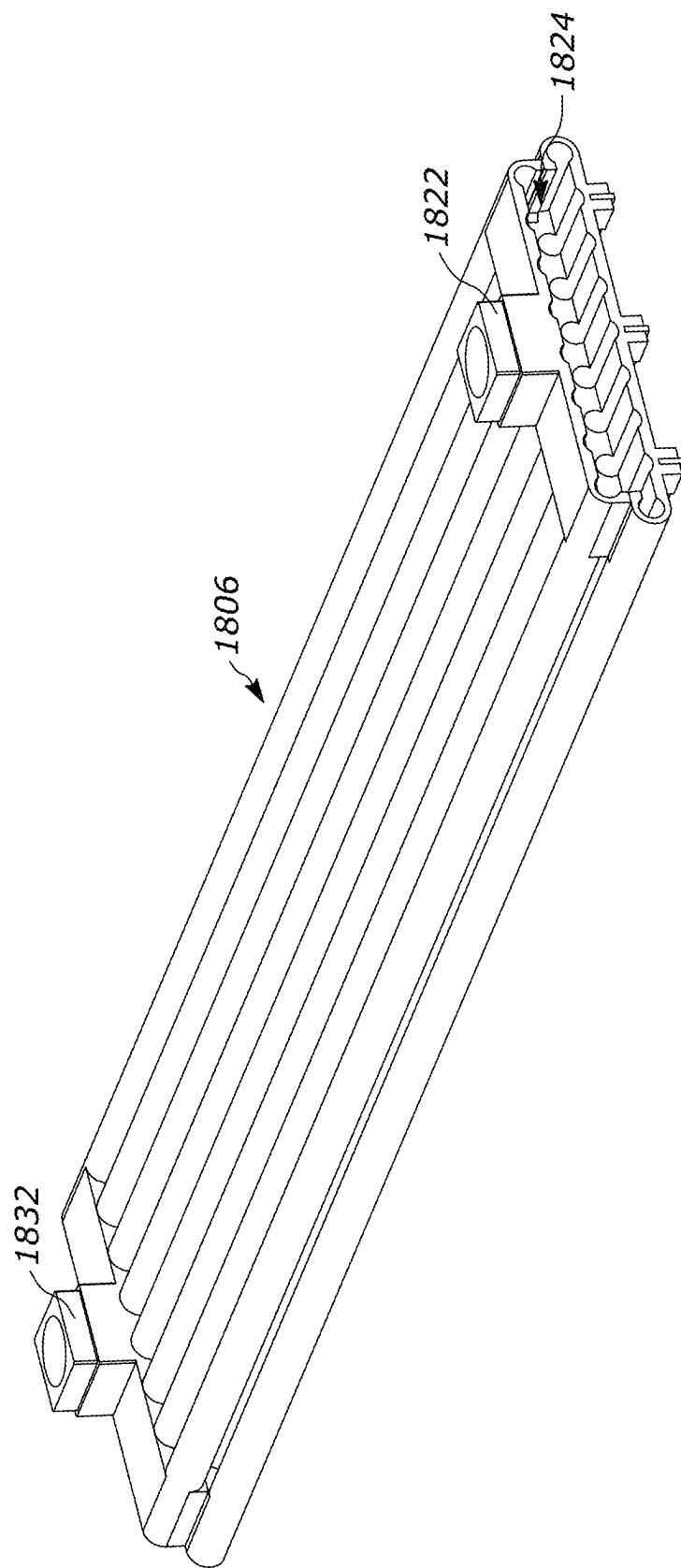


FIG. 18B

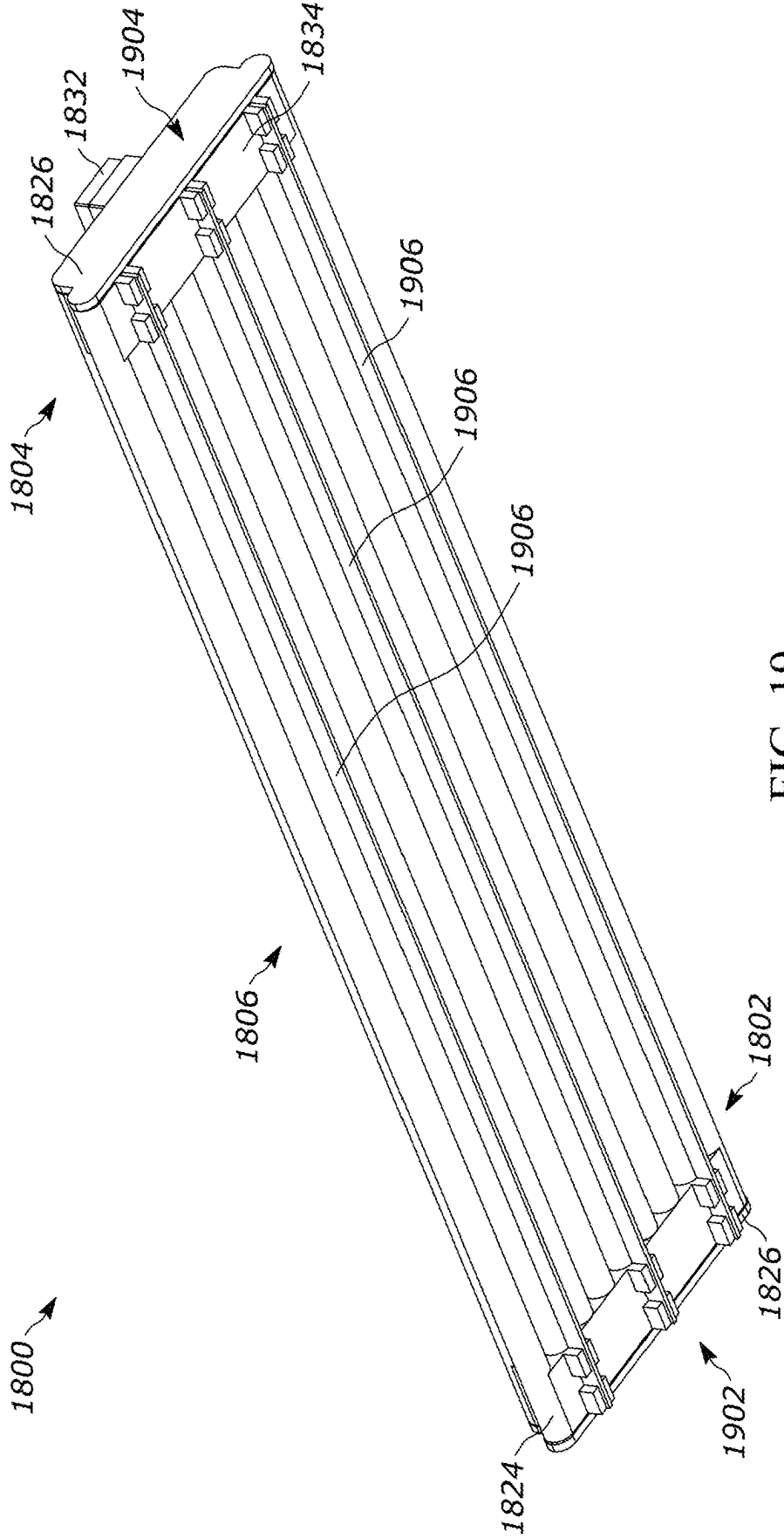


FIG. 19

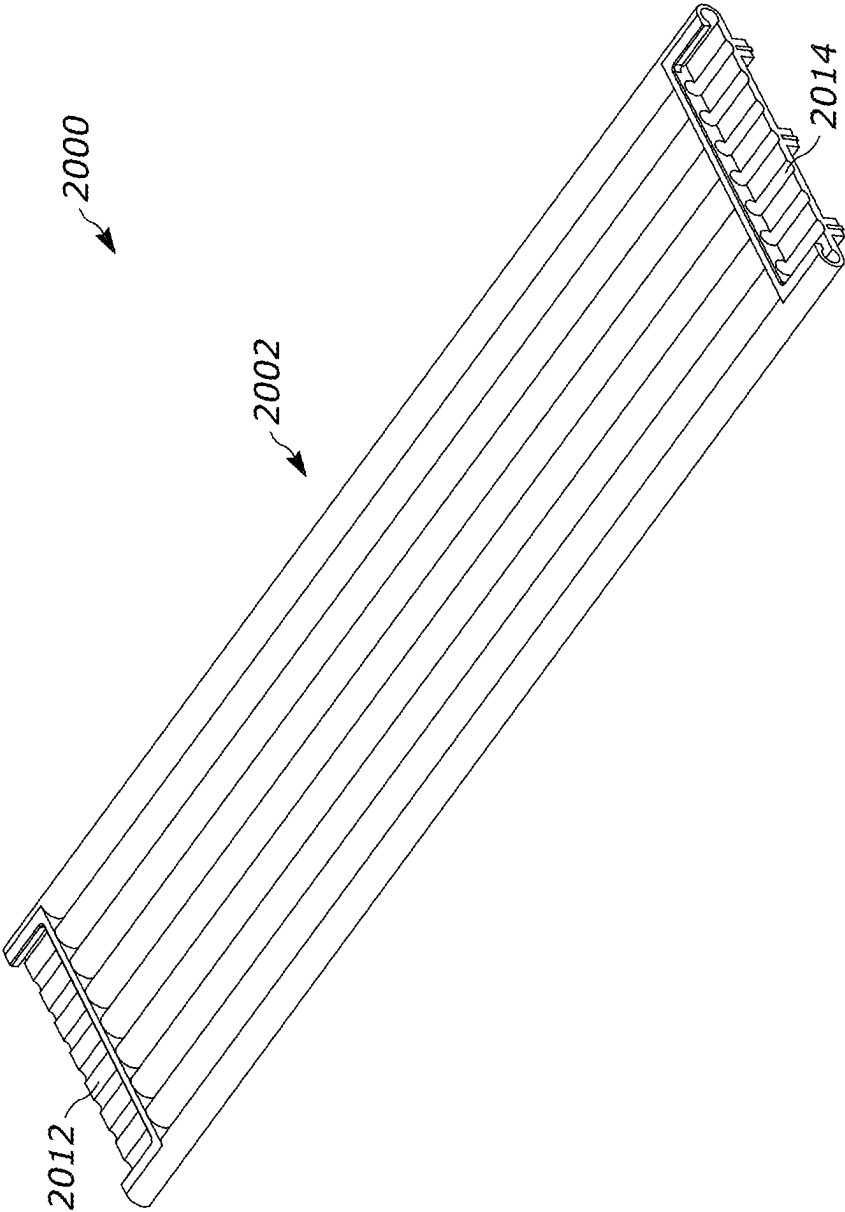


FIG. 20

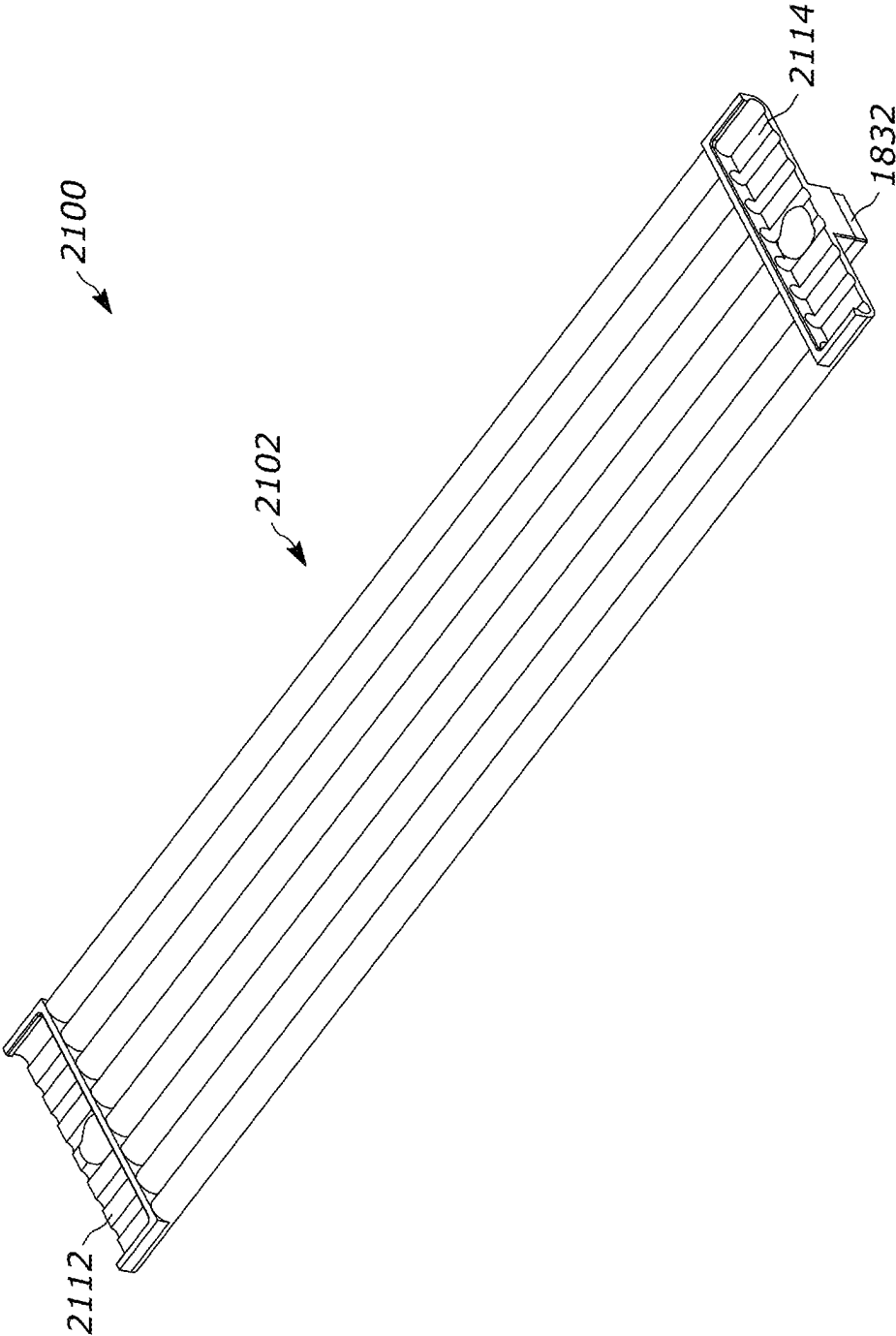


FIG. 21

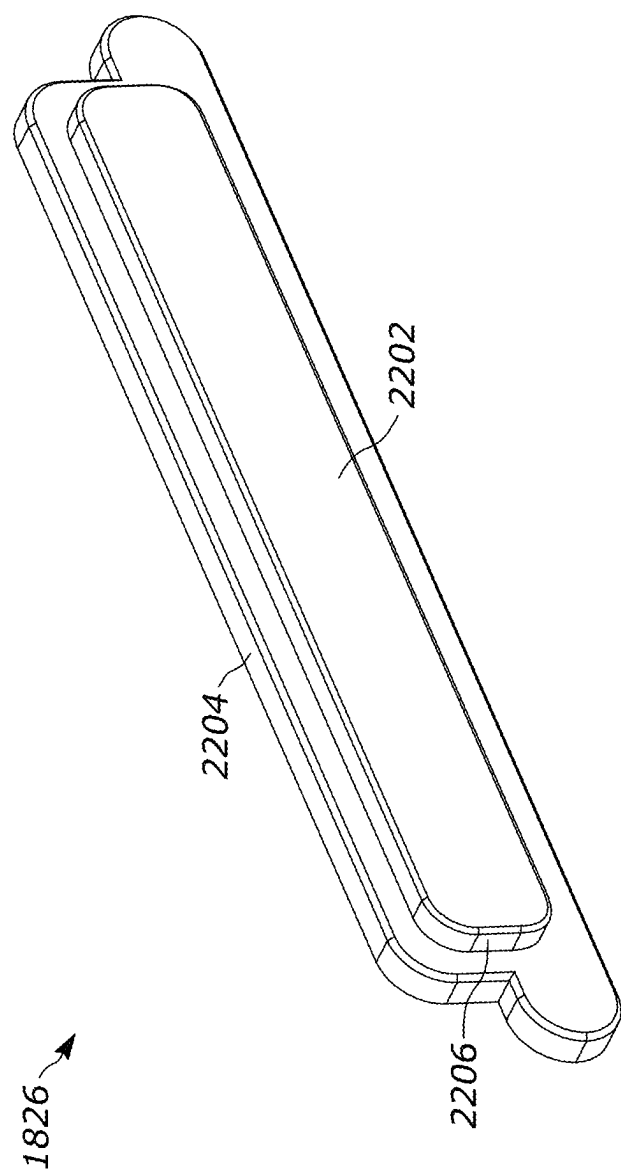


FIG. 22

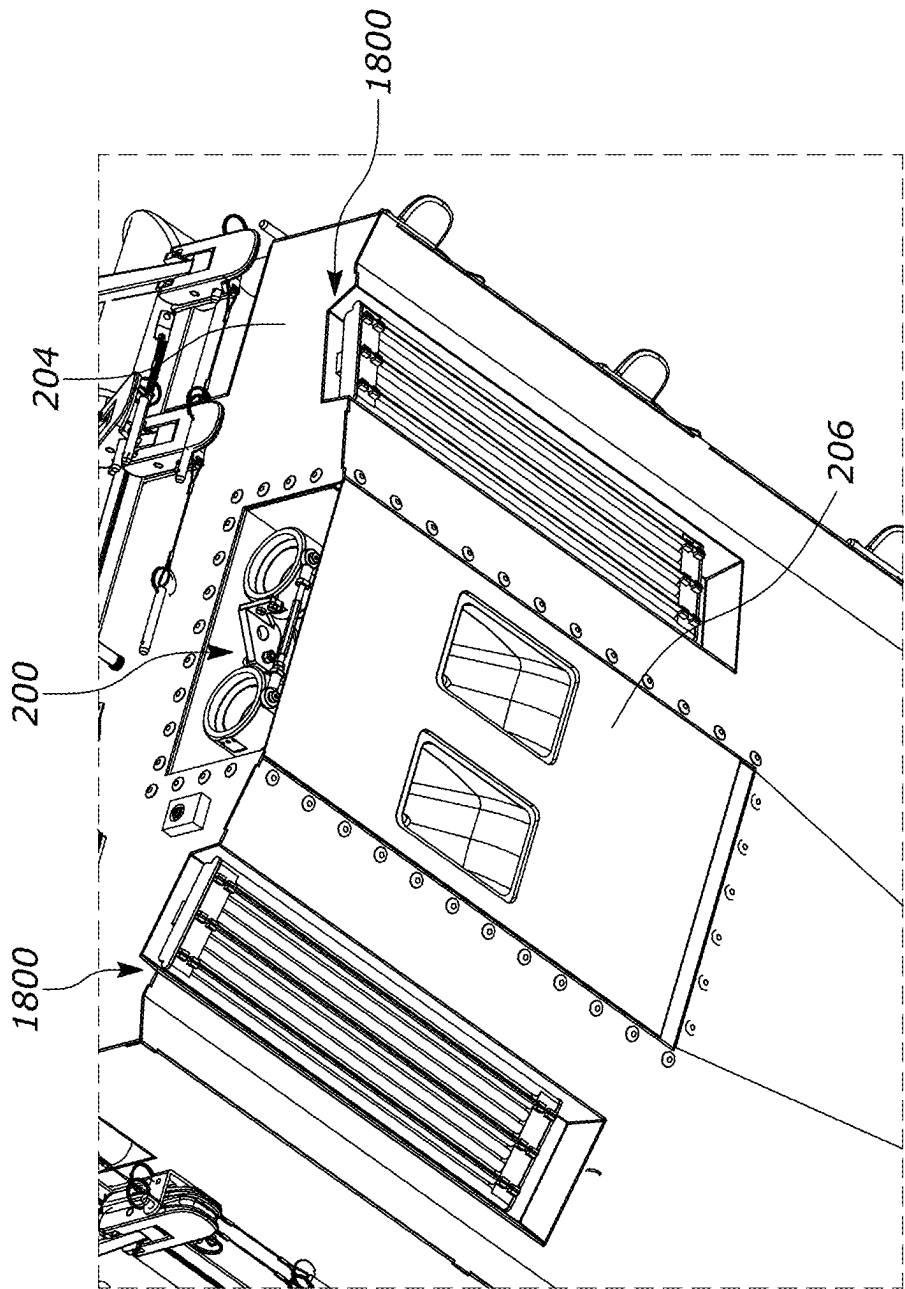


FIG. 23

MICRO-KEEL COOLER FOR MARITIME VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to cooling systems and, more particularly, to a micro-keel cooler for a maritime vehicle.

BACKGROUND

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

[0004] Maritime vehicles generally utilize thrust systems for causing movement of the maritime vehicle. In some cases, maritime vehicles utilize propeller-based thrust systems. Maritime vehicles can alternatively utilize jet-based pump thrust systems, whereby propellers extend outside the hull of the maritime vehicle while jet pumps pass water through a channel and through an impeller disposed within the maritime vehicle to generate thrust. The jet pump forces the water through a directional nozzle to control the direction of thrust. As a result, the jet pump controls the forward speed and direction of the maritime vehicle.

[0005] Additionally, maritime vehicles require various cooling systems to expel the heat generated in operating the vehicle. For example, thrust systems generally include internal combustion or electrical motors that generate large amounts of thermal energy. Cooling systems absorb heat from the thrust system and transfer the thermal energy into the surrounding environment. In some examples, maritime vehicles can incorporate heat exchangers that direct the thermal energy into the surrounding water.

SUMMARY

[0006] A micro-keel cooler for cooling a coolant flowing through a heat-generating component of a maritime vehicle, and a cooling system including one or more micro-keel coolers. The cooler includes an inlet adapted to receive the coolant at a first temperature and an outlet adapted to output the coolant at a second temperature less than the first temperature. The cooler further includes a plurality of tubes fluidly coupled to an inlet mixing chamber of the inlet and an outlet chamber of the outlet and configured to direct the coolant from the inlet mixing chamber to the outlet mixing chamber, and a plate coupled to the inlet and the outlet and disposed adjacent to the tubes, the plate configured to direct a fluid over at least a portion of the tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a cooling temperature that is less than the first temperature.

BRIEF DESCRIPTION OF DRAWINGS

[0007] The present disclosure is described in the following detailed description in conjunction with the drawings, wherein:

[0008] FIG. 1A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure.

[0009] FIG. 1B is a front view of the maritime vehicle of FIG. 1A.

[0010] FIG. 1C is a rear view of the maritime vehicle of FIG. 1A.

[0011] FIG. 1D is a bottom perspective view of the maritime vehicle of FIG. 1A.

[0012] FIG. 1E is another bottom perspective view of the maritime vehicle of FIG. 1A.

[0013] FIG. 1F is similar to FIG. 1A, but with the cap and various components of the maritime vehicle removed for illustrative purposes.

[0014] FIG. 1G is similar to FIG. 1F, but with additional components of the maritime vehicle removed for illustrative purposes.

[0015] FIG. 1H is a rear view of the maritime vehicle of FIG. 1F.

[0016] FIG. 1I is a first cross-sectional view taken along line I-I in FIG. 1A.

[0017] FIG. 1J is a second cross-sectional view taken along line J-J in FIG. 1A.

[0018] FIG. 1K is similar to FIG. 1A, but with the latching assembly of the maritime vehicle removed for illustrative purposes.

[0019] FIG. 2 is a front perspective view of an example dual jet pump assembly made in accordance with the present disclosure and usable in connection with the maritime vehicle of FIGS. 1A-1K.

[0020] FIG. 3 is a perspective view of the jet pump drive systems of the jet pump assembly of FIG. 2.

[0021] FIG. 4 is a perspective view of the motors of the jet pump drive systems of FIGS. 2 and 3.

[0022] FIG. 5 is a perspective view of the drive shaft and impeller of one of the jet pump drive systems of FIGS. 2 and 3.

[0023] FIG. 6 is a perspective view of the jet pump nozzle of one of the jet pump drive systems of the jet pump assembly of FIG. 2.

[0024] FIG. 7 is a perspective view of the transom assembly of the jet pump assembly of FIG. 2.

[0025] FIG. 8 is a perspective view of one of the transom housings of the transom assembly of FIG. 7.

[0026] FIG. 9 is a side view of the transom housing of FIG. 8.

[0027] FIG. 10A is a front, perspective view of the transom plate of the transom assembly of FIG. 7.

[0028] FIG. 10B is a rear plan view of the transom plate of FIG. 10A.

[0029] FIG. 11 is a front, perspective view of the transom box of the transom assembly of FIG. 7 and portions of the jet pump drive systems of FIGS. 2 and 3.

[0030] FIG. 12 is a rear perspective view of FIG. 2.

[0031] FIG. 13A is a front perspective view of the bottom plate of the dual jet pump assembly of FIG. 2.

[0032] FIG. 13B is a rear perspective view of FIG. 13A.

[0033] FIG. 14 is a perspective bottom view of the bottom plate of the dual jet pump assembly of FIG. 2.

[0034] FIG. 15 is a perspective view of the jet pump assembly of FIG. 2 installed in the maritime vehicle of FIGS. 1A-1K.

[0035] FIG. 16 is a bottom view of the jet pump assembly of FIG. 2 installed in the maritime vehicle of FIGS. 1A-1K.

[0036] FIG. 17 is a schematic diagram of one example of a cooling system made in accordance with the present disclosure and usable to cool the jet pump assembly of FIG. 2.

[0037] FIG. 18A is a perspective top view of one example of a micro-keel cooler made in accordance with the present disclosure and usable in the cooling system of FIG. 17.

[0038] FIG. 18B another perspective top view of the micro-keel cooler of FIG. 18A but with an end plate removed.

[0039] FIG. 19 is a perspective bottom view of the micro-keel cooler of FIGS. 18A and 18B.

[0040] FIG. 20 is a perspective view of a bottom portion of the micro-keel cooler of FIGS. 18A and 18B.

[0041] FIG. 21 is a perspective view of a top portion of the micro-keel cooler of FIGS. 18A and 18B.

[0042] FIG. 22 is a perspective view of one of the end plates of the micro-keel cooler of FIGS. 18A and 18B.

[0043] FIG. 23 is a close-up, bottom perspective view of a portion of the maritime vehicle of FIGS. 1A-1K and including two of the micro-keel coolers of FIGS. 18A and 18B.

[0044] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

[0045] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0046] FIGS. 1A-1K illustrate one example of a maritime vehicle 100 constructed in accordance with the teachings of the present disclosure. The maritime vehicle 100 is an unmanned vessel configured to autonomously traverse a body of water. The maritime vehicle 100 generally includes a hull 104 and a cap 108 that is coupled to the hull 104 to secure various components within the maritime vehicle 100. The hull 104 is at least partially disposed in the body of water in which the maritime vehicle 100 is traversing. The hull 104 in this example is a mono-hull that has a front (or bow) 112, a rear (or stern) 116, two sides 120, and a keel 124 coupled to another. The front 112, the rear 116, the sides 120, and the keel 124 can be welded together or can be coupled to one another in a different manner. For example, the front 112, the rear 116, the sides 120, and the keel 124 can be coupled together in the manner described in U.S. Provisional Application No. 63/561,282, titled "Systems and Approaches for Assembling a Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The hull 104 is configured such that the hull provides a continuous planing surface that allows the maritime vehicle 100 to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap 108 is coupled to the hull 104 to cover and/or conceal the components of the maritime vehicle 100 disposed in and carried by the hull 104 as the maritime vehicle 100 traverses the body of water.

[0047] In this example, the hull 104 and the cap 108 each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull 104 is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle 100 can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull. In this example, the cap 108 entirely covers the hull 104 (and the components therein). In other examples, however, the maritime vehicle 100 need not include the cap 108 or the cap 108 may only partially cover the hull 104 (and the components disposed therein).

[0048] In some examples, the cap 108 can be removably coupled to the hull 104 via a locking system. For example, as illustrated in FIGS. 1A-1K, the locking system 128 can take the form of a plurality of latch mechanisms 128 disposed around a perimeter of the maritime vehicle 100. Thus, the cap 108 can be removed to allow access to the interior of the hull 104. In other examples, however, the cap 108 can be permanently coupled to the hull 104 to permanently conceal the components within the maritime vehicle 100.

[0049] The maritime vehicle 100 also includes a plurality of bulkheads 132 arranged within the hull 104. The bulkheads 132 divide the maritime vehicle 100 into a plurality of different compartments for receiving and retaining different components in the maritime vehicle 100.

[0050] The maritime vehicle 100 also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle 100 as well as data about the environment surrounding the maritime vehicle 100 (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle 100. The sensors can include, for

example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle 100, the pressure of the water in which the maritime vehicle 100 is disposed, the pressure within the maritime vehicle 100), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle 100, a temperature of ambient air external to the maritime vehicle 100, a temperature of water in which the maritime vehicle 100 is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0051] The maritime vehicle 100 also includes a power system that is generally configured to power the maritime vehicle 100 (and the components of the maritime vehicle 100). The power system generally includes a thrust system and one or more power sources configured to power the thrust system (and the other components within the maritime vehicle 100). The thrust system is generally configured to propel the maritime vehicle 100 in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle 100, hydrogen stored in hydrogen tanks carried by the maritime vehicle 100, solar panels (e.g., mounted to an exterior of the maritime vehicle 100), or other sources. The maritime vehicle 100 illustrated in FIGS. 1A-1K includes four battery assemblies each including a rechargeable battery. The maritime vehicle 100 illustrated in FIGS. 2A-2K also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled "Power System for Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The maritime vehicle 100 generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle 100.

[0052] In operation, the maritime vehicle 100 may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull 104, attached to the exterior surface of the hull 104, or attached to the exterior surface of the cap 108 prior to deployment. Likewise, retrieved payloads can be stored in the hull 104, attached to and stored on the exterior surface of the hull 104, or attached to and stored on the exterior surface of the cap 108.

[0053] The maritime vehicle 100 can also include other systems to help with the operation of the maritime vehicle 100, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle 100 in the water, regardless of whether the maritime vehicle 100 is stationary or on the move. To this end, the maritime vehicle 100 may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle 100. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle 100. The ballast system may also provide for the selective submerging and resurfacing of the maritime vehicle 100 in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle 100 and to guide the maritime vehicle 100 to its desired location in the body of water (or in a different body of water). The vision system is generally configured to capture, process, and analyze images obtained by one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle 100 (including objects in that environment).

[0054] The maritime vehicle 100 further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle 100 and one or more central (remote) controllers, (ii) between the maritime vehicle 100 and and/or one or more other maritime vehicles 100 and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle 100. The communications system generally includes one or more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle 100 and the central controller(s) or other maritime vehicles 100. For example, the maritime vehicle 100 includes a plurality of antennae disposed on an exterior of the cap 108 as well as a plurality of antennae disposed in the hull 104.

[0055] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles 100 or military assets) and to perform automated operations of the maritime vehicle 100 based on that data. In some examples, the maritime vehicle 100 includes a plurality of different local controllers. For example, the maritime vehicle 100 can include one or more thrust controllers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0056] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle 100 includes several additional components. For example, the maritime vehicle 100 includes various sealing elements configured to provide seals between different components of the maritime vehicle 100 (or between the maritime vehicle 100 and the environment surrounding the maritime vehicle 100). As another example, the maritime vehicle 100 also includes various fasteners that help to couple the components of the maritime vehicle 100 together. As yet another example, the maritime vehicle 100 includes cabling that helps to communicatively couple components of the maritime vehicle 100 together. As yet another example, the maritime vehicle 100 includes various electrical components that help to operate the maritime vehicle 100, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, one or more brain boards.

[0057] Further described herein is a compact and lightweight jet pump assembly and micro-keel cooler system for a small maritime vehicle (e.g., the maritime vehicle 100). One example of such a compact and lightweight jet pump assembly 200 is illustrated in FIGS. 2-16. In this example, the jet pump assembly includes two jet pumps in a single assembly. In other examples, the jet pump assembly can include one jet pump or more than two jet pumps. Additionally, each of the jet pumps is preferably coupled to a cooling system such as the cooling system described in connection with FIG. 17 or the micro-keel cooler system, one example of which is illustrated in and described in connection with FIGS. 18-22. In some examples, a single micro-keel cooler system can be coupled to two or more jet pumps.

Jet Pump Assembly

[0058] Jet pump assemblies generally include a water intake, an impeller, a diffuser, a nozzle, and a steering system. Overall, jet pump assemblies offer a compact and efficient means of propulsion and steering for small maritime vehicles (e.g., small boats and small vessels) and jet skis, making them well-suited for use in shallow water and areas where traditional propellers may be impractical or unsafe.

[0059] The dual jet pump assembly 200 illustrated in FIGS. 2-16 is made in accordance with the present disclosure and is configured for use in a maritime vehicle such as the maritime vehicle 100. As illustrated in FIG. 2, the dual jet pump assembly 200 includes two jet pump drive systems 202 (described in greater detail in connection with FIGS. 3-5), a transom assembly 204 (described in greater detail in connection with FIGS. 7-12), and a bottom plate 206 (described in greater detail below in connection with FIGS. 13A, 13B, and 14). The dual jet pump assembly 200 is manufactured as a functional unit that can be assembled prior to installation in the maritime vehicle (and easily and quickly removed from the maritime vehicle when replacement or repair is necessary). For example, the dual jet pump assembly 200 can be manufactured and bench tested prior to installation in the maritime vehicle. While in the present example, the dual jet pump assembly 200 includes two jet pump drive systems 202 and the transom assembly 204, configured for the two jet pump drive systems 202, in other examples, the jet pump assembly can include one jet pump drive system or more than two jet pump drive systems.

Jet Pump Drive System

[0060] FIGS. 3-6 illustrate the jet pump drive systems 202 of FIG. 2 in greater detail. As illustrated, each jet pump drive system 202 includes a motor 302 (shown in detail in FIGS. 3 and 4), a drive shaft 306 (shown in detail in FIGS. 3 and 5), an impeller 308 (shown in detail in FIGS. 3 and 5), and a nozzle 602 (shown in FIG. 6). Each jet pump drive system 202 is identical to the other, but in some examples, the jet pump drive systems 202 can be mirrored or otherwise altered to meet the functional requirements of the maritime vehicle (e.g., maritime vehicle 100). In the present example, the motor 302 is coupled to the drive shaft 306 via a coupling 312 and the impeller 308 is coupled to (e.g., integrally formed with) the drive shaft 306. In other examples, the drive shaft 306 can be integrally formed with the motor 302 and/or the impeller 308 can be coupled to the motor 302 via a coupling similar to the coupling 312. Each component of the jet pump drive system 202 will be described in greater detail below.

[0061] The motors 302 are electric motors and can have any functional number of phases or poles. Each motor 302 includes various electrical power and control connections 402. Each motor 302 may include a motor controller (e.g., an electronic speed controller (ESC)), not shown, to control the speed of the motor 302. The motor controller can be integrated in the motor 302 or can be disposed separate from and in electric communication with the motor 302. In other examples, however, the motors 302 can be any type of motor, engine, or power source (e.g., internal combustion motor). Additionally, because in the present example each jet pump drive system 202 includes one motor 302 for each drive shaft 306 and impeller 308, the two motor controllers can independently control the two motors 302 (and the two drive shafts 306 and impeller 308 respectively coupled thereto), respectively. In other examples, however, a single motor 302 can be configured to actuate both drive shafts 306. In such examples, the single motor 302 can be coupled to the drive shafts 306 via a gear assembly.

[0062] During operation, each motor 302 generates excess thermal energy and can get quite hot. Accordingly, the motors 302 often require cooling. As shown in FIG. 4, each motor 302 in this example includes a cooling jacket 412 and couplings 422, 424 that couple the cooling jacket 412 to a cooling system such as the cooling system described in greater detail in connection with FIG. 17. In other examples, the motor 302 may be air cooled or temperature regulated using another cooling method. In such examples, the motor 302 may not include the cooling jacket 412 and/or the couplings 422, 424.

[0063] As shown in FIG. 5, the drive shaft 306 includes a first end 502 and a second end 504. The first end 502 couples to the motor 302 via the coupling 312 and the second end 504 couples to the impeller 308. The drive shaft 306 is configured to transmit rotational energy from the motor 302 to the impeller 308. As a result, the drive shaft 306 is rigid and capable of withstanding the torque of the motor 302. In various examples, the drive shaft 306 can be made of a strong metal, polymer, or similar material. The drive shaft 306 is cylindrical, but in other examples, the drive shaft 306 can have any other cross-sectional shape that is radially symmetrical and capable of rotating at high revolutions per minute.

[0064] The drive shaft 306 also includes a slot 512, near the first end 502, for facilitating the connection with the

coupling 312. The slot 512 may engage with a set screw of the coupling 312 or include a shoulder to releasably couple the coupling 312 to the slot 512. Additionally, in the present example, the drive shaft 306 is coupled to the impeller 308 at the second end 504 via a coupling 514. In another example, the second end 504 of the drive shaft 306 can be inserted into an opening of the impeller 308 and welded to the impeller 308. In yet other examples, the drive shaft 306 can be coupled to the impeller 308 via a different but known method of coupling rotating components. Likewise, the drive shaft 306 can be coupled to the motor 302 in any number of different known manners. For example, the drive shaft 306 can be welded to the motor 302.

[0065] As also shown in FIG. 5, the impeller 308 includes impeller blades 552 disposed on an impeller head 554. The impeller head 554 includes a bearing 556 for coupling to the nozzle 602 (discussed in greater detail in connection with FIG. 6). In this example, the impeller blades 552 are symmetrically disposed about the head 554 to ensure stability during operation. During operation, the impeller head 554 and the impeller blades 552 are rotated about a central axis 560 and impart momentum to a fluid (e.g., water) in the direction 562. As a result, the impeller 308 increases the speed of the fluid in the direction 562.

[0066] The blades 552 of the impeller 308 are curved and shaped to improve the effectiveness of the impeller 308. In the present example, the impeller 308 includes four blades 552, but in other examples, the impeller 308 can include more or fewer blades 552. Additionally, a radial length 564 of the blades 552 can be selected so as to efficiently operate in the fluid. For example, the radial length 564 of the blades 552 can be limited to reduce the possibility of cavitation in water.

[0067] FIG. 6 illustrates one of the nozzles 602 that may couple to the impellers 308. Each nozzle 602 generally includes a diffuser 604, a nozzle 606, and a directional outlet 608. The diffuser 604 gradually transitions the flow area of the nozzle 606 to create a reactive force to generate a propulsive force. The diffuser 604 includes a tail cone 612 including an aperture 614 sized to receive the bearing 556. The directional outlet 608 of the nozzle 602 is configured to direct the propulsive force of the fluid exiting the nozzle 602, thereby permitting control of the direction of movement of the maritime vehicle (e.g., the maritime vehicle 100). In the present example, the directional outlet 608 is pivotable about the pivoting axis 622. In other examples, the directional outlet 608 can be actuable about a different axis or two axes.

Transom Assembly

[0068] The transom assembly 204 of the jet pump assembly 200 provides an enclosed space that receives water from the bottom of the maritime vehicle 100 and, through operation of the jet pump drive systems 202 of FIGS. 2-6, expels high-speed water from the aft of the vehicle to propel the vehicle forward. The transom assembly 204 generally includes two transom housings 702 (one for each jet pump drive system 202 and described in greater detail in connection with FIGS. 8 and 9), two transom plates 704 (described in greater detail in connection with FIGS. 10 and 11), and a transom box 706 (described in greater detail in connection with FIG. 12). The transom assembly 204 is configured to be coupled with the jet pump drive systems 202 via apertures 712, respectively. For example, the drive shafts 306 pass

through the apertures 712, as best illustrated in FIG. 2. In some examples, water-tight seals are utilized to keep the apertures 712 watertight.

[0069] As shown in FIGS. 8 and 9, each transom housing 702 includes a bottom coupling 802, an aft coupling 804, and a transom channel 806 disposed between the bottom coupling 802 and the aft coupling 804. The bottom coupling 802 of each transom housing 702 is configured to engage the bottom plate 206 as shown in FIG. 2 and described in greater detail in connection with FIGS. 13A & 13B. Meanwhile, the aft coupling 804 of each transom housing 702 is configured to engage the transom plate 704 as shown in FIGS. 2 and 7. The transom channel 806 is disposed and extends between the bottom coupling 802 and the aft coupling 804 and serves as the pump intake for each respective jet pump drive system 202.

[0070] As shown in FIG. 9, the bottom coupling 802 of each transom housing 702 includes a coupling plate 822 and a lower plate 824. The coupling plate 822 is provided to fasten the respective transom housing 702 to the bottom plate 206. In some examples, the coupling plate 822 can be bolted, riveted, or welded to the bottom plate 206. The lower plate 824 is configured to be flush with the exterior of a hull (e.g., the hull 104) of the maritime vehicle when the transom housing 702 and bottom plate 206 are installed on the vehicle. In some examples, the lower plate 824 can further include a protective cover or grate to limit debris entering the transom housing 702.

[0071] Further, the aft coupling 804 of each transom housing 702 includes a ring 832 and a flange 834. The ring 832 passes through a respective aperture in the transom plate 704 (described in greater detail below in connection with FIGS. 10 and 11). In some examples, the nozzles 602 are coupled to the rings 832, respectively. The rings 832 pass through the transom plate 704 but the flange 834 of each transom housing 702 is coupled to the transom plate 704 via rivets, fasteners, and/or welding.

[0072] The transom channel 806 of each transom housing 702 is disposed between the bottom coupling 802 and the aft coupling 804. The transom channel 806 provides a smooth transition from the bottom coupling 802 to the aft coupling 804. As shown in FIG. 9, the aperture 712 of each transom housing 702 is axially aligned with the respective aft coupling 804. As a result, the drive shafts 306 and the impellers 308 can pass through the apertures 712, respectively, and, in turn, be automatically centered in the respective aft coupling 804 about an axis 960 that is coaxially aligned with the respective central axis 560.

[0073] Turning now to FIGS. 10 and 11, the transom plate 704 is provided to ensure proper alignment of the jet pump drive systems 202 and the transom assembly 204 with the bottom plate 206, as shown in FIG. 2. The transom plate 704 includes alignment apertures 1002. In some examples, fasteners (not shown) can be disposed in the alignment apertures 1002 (and corresponding apertures formed in the bottom plate 206) to align the transom plate 704 with the bottom plate 206 such that the transom plate 704 is aligned with the jet pump drive systems 202. In turn, at least in this example, the transom plate 704 is disposed perpendicular to the bottom plate 206. By auto-aligning the transom plate 704 with the other components of the jet pump assembly 200, the jet pump assembly 200 is easier to assemble and can be assembled apart from the maritime vehicle (e.g., the maritime vehicle 100).

[0074] The transom plate 704 includes impeller apertures 1004 and a steering aperture 1006. The impeller apertures 1004 are sized and arranged to receive the impellers 308, respectively. In the present example, the jet pump assembly 200 includes one steering assembly (not shown) configured to control the direction of both jet pump drives 202 via the single steering aperture 1006. In other examples, the transom plate 704 can include more steering apertures 1006 to accommodate additional steering assemblies.

[0075] As shown in FIG. 10A, the transom plate 704 includes a plurality of apertures 1020 for receiving various fasteners (e.g., apertures to secure the transom channel 702 to the transom plate 704). In the present example, apertures 1020 are circumferentially arranged about each of the impeller apertures 1004. Each of the plurality of apertures 1020 is a blind aperture (i.e., is not visible on the front surface 1052 or, more generally, from the front of the transom plate 704, as shown in FIG. 10B). As a result, none of the plurality of apertures 1020 can be a source of water leakage. Additionally, by reducing the amount of manufacturing time required to seal each of the plurality of apertures 1020, the manufacturing time for the jet pump assembly 200 is reduced.

[0076] The transom plate 704 is coupled to the transom box 706 to form a watertight enclosure. In the present example, the front surface 1052 of the transom plate 704 includes slots 1054 that are disposed about the perimeter of the front surface 1052 and are configured to couple the transom plate 704 to the transom box 706.

[0077] As illustrated in FIG. 11, the transom box 706 includes a top wall 1202 and two side walls 1204 coupled to the top wall 1202. Each of the top and side walls 1202, 1204 includes tabs 1208 that are inserted into the slots 1054 disposed in the front surface 1102 of the transom plate 704 so as to couple the transom box 706 to the transom plate 704. The transom box 706 also includes a lip 1212 that extends outward (upward in the orientation shown in FIG. 11) from the top and side walls 1202, 1204. The lip 1212 can be sealed against an interior surface of the hull (e.g., the hull 104) of the maritime vehicle. The lip 1212 can, for example, be welded, riveted, or otherwise fastened to the hull. As shown in FIG. 15, the lip 1212 is fastened to the hull via bolts.

[0078] Finally, FIG. 12 illustrates the transom box 706 when coupled to the transom housings 702, the transom plate 704, and the bottom plate 206. The transom box 706, along with the transom plate 704 and the bottom plate 206, form an enclosure open to the outside environment of the maritime vehicle 100. In the present example, at least part of the impeller 308 and the nozzle 602 are disposed in the transom box 706 (and the enclosure defined by the transom box 706).

Bottom Plate

[0079] FIGS. 13A & 13B illustrate the bottom plate 206 in greater detail. The bottom plate 206 is configured to secure and align the jet pump drive system 202 and the transom assembly 204. In the present example, the bottom plate 206 is a single piece of metal that includes a base 1300, a plurality of apertures 1302, and a plurality of pockets. The apertures 1302 are disposed in the base 1300 to help align and secure the components of the jet pump drive system 202 and the transom assembly 204, whereas the pockets are sized and arranged in the base 1300 to receive various components of the jet pump drive system 202 and the transom assembly

204. In other examples, the bottom plate 206 can be made of two or more sheets of metal fastened or welded together.

[0080] In the present example, the bottom plate 206 includes five pockets—pockets 1312, 1314, and 1316—formed in the base 1300. The first and second pockets 1312 of the bottom plate 206 are sized to receive and retain the motors 302. The third and fourth pockets 1314 include openings 1315 that extend through the bottom plate 206 and are sized to receive and be enclosed by the bottom coupling 802 of the transom housings 702, respectively, when the coupling plates 822 of the transom housing 702 are seated against the portions of the pockets 1314 surrounding the openings 1315, respectively. Finally, the fifth pocket 1316 is sized to receive the transom plate 704 and the transom box 706.

[0081] As shown in FIGS. 13A, 13B, and 14, the base 1300 of the bottom plate 206 has an upper surface 1400 and a lower surface 1402 opposite the upper surface 1400. As best shown in FIG. 14, the bottom plate 206 also includes a lip 1404 offset from the lower surface 1402 of the base 1300 by a distance 1406. The lip 1404 is offset from the lower surface 1402 so that the lower surface 1402 can be flush with an external surface of the hull of the maritime vehicle. As a result, the distance 1406 is approximately equal to the thickness of the hull. In turn, when the lip 1404 is coupled to an interior surface of the hull, the lower surface 1402 can help create a flush, uniform exterior surface of the maritime vehicle. By creating a flush, uniform exterior surface, the lower surface 1402 reduces drag and reduces the likelihood of generating cavitation along the exterior surface of the hull.

[0082] The lip 1404 generally extends along three of the four sides of the bottom plate 206. In some examples, the bottom plate 206 also includes a vertical lip 1408 that extends from the lip 1404. The vertical lip 1408 is configured to abut the lip 1212 of the transom box 706. Preferably, the lip 1408 and the lip 1212 can be sealed to one another with welding and/or marine-grade sealant to make a watertight seal between the transom box 706 and the bottom plate 206. In other examples, a seal can be provided between the lip 1212 and the lip 1404.

Assembly and Installation

[0083] In FIG. 2, the dual jet pump assembly 200 is fully assembled, and FIG. 15 illustrates the fully assembled dual jet pump assembly 200 installed in the maritime vehicle 100. Beneficially, the dual jet pump assembly 200 can be fully assembled apart from the maritime vehicle 100. In other examples, the dual jet pump assembly 200 can be simultaneously assembled and installed on the maritime vehicle 100. When assembling the dual jet pump assembly 200, either the motors 302 or the transom assembly 204 can be installed on the bottom plate 206 first. After the transom assembly 204 and the motors 302 are installed on the bottom plate 206, the rest of the jet pump drive system 202 can be installed.

[0084] The transom assembly 204 is coupled to the bottom plate 206 by first coupling the transom plate 704 to the bottom plate 206. As discussed above, the transom plate 704 includes the alignment apertures 1002 to ensure the transom plate 704 is properly aligned with the bottom plate 206, and fasteners (e.g., pins, not shown) can be disposed in the alignment apertures 1002 and alignment apertures 1322 formed in the bottom plate 206 (see FIG. 13B) and aligned

with the alignment apertures **1002** when the transom plate **704** is properly aligned with the bottom plate **206**. Next, the transom box **706** is coupled to the transom plate **704** and the bottom plate **206**. In various examples, the lip **1212** of the transom box is aligned with the lip **1404** of the bottom plate **206** and the edge of the lip **1212** is sealed to the edge of the lip **1404**. The transom housings **702** are then coupled to the bottom plate **206** and the transom plate **704**.

[0085] To couple the motors **302** to the bottom plate **206**, motor brackets **212** are installed on the bottom plate **206** (see FIG. 15). In some examples, the brackets **212** include alignment pins (not shown) to ensure the brackets **212** are properly placed and aligned on the bottom plate **206**. The motors **302** can then be secured in the motor brackets **212**.

[0086] With the motors **302** and the transom assembly **204** installed on the bottom plate **206**, the drive shafts **306**, the impellers **308**, and the nozzles **602** can be installed. In the present example, the first end **502** of each drive shaft **306** is passed through the transom assembly **204**, through the aperture **712** of a respective one of the transom housings **704**, and coupled to a respective one of the motors **302** via the respective coupling **312**.

[0087] At this point, the jet pump assembly **200** is functionally assembled. The jet pump assembly **200** can then be sealed so as to be watertight. In various examples, joints between the bottom plate **206** and the transom assembly **204** can be welded or sealed with marine-grade sealants. With the components of the jet pump assembly **200** functionally assembled and fully sealed, the jet pump assembly **200** can be bench tested and installed on the vehicle.

[0088] As shown in FIGS. 15 and 16, the jet pump assembly **200** is bolted to the hull **104** of the maritime vehicle **100**. In other examples, the jet pump assembly **200** can be rivetted or welded to the hull **104**. Installing the jet pump assembly **200** can be done entirely from the inside of the maritime vehicle **100**. Unlike other jet pump assemblies, the jet pump assembly **200** does not require any underside access to the maritime vehicle **100**. Furthermore, because the jet pump assembly **200** is already functional and properly assembled, the jet pump assembly **200** does not require extensive alignment and testing after installed in the maritime vehicle **100**. Additionally, in some examples, the weight of the jet pump assembly **200** is low enough that the jet pump assembly **200** can be fully assembled, tested, and installed by a single person.

[0089] The jet pump assembly **200** is lowered into the opening at the aft, bottom portion of the maritime vehicle **100**. The lip **1212** and the lip **1404** are inserted against the interior surface of the hull **104**. In some examples, the lips **1212**, **1404** are riveted, bolted (or otherwise fastened), or welded to the hull **104**. In some examples, the jet pump assembly **200** further requires marine sealant to prevent or reduce water leaks between the jet pump assembly **200** and the hull **104** of the maritime vehicle **100**.

[0090] After installing the jet pump assembly **200** to the maritime vehicle **100**, the jet pump assembly **200** can be coupled to the various other systems and components of the maritime vehicle. For example, the motors **302** can be electrically connected to a power source (not shown) and the electronic controller(s) (not shown). In some examples, the power source is a battery, generator, or other electrical power source. Additionally, the jet pump assembly system

200 can be functionally connected to a vehicle cooling system (e.g., cooling system **1700** described in greater detail in connection with FIG. 17).

Operation of the Jet Pump Assembly

[0091] During operation, water first enters the water intake of the maritime vehicle **100** and proceeds through the transom channels **806** to the impellers **308**. The water intake generally draws water into the transom channels **806** through an opening or grate in the bottom of the maritime vehicle **100**, e.g., the openings **1315**. In some examples, the openings **1315** and transom channels **806** are fully submerged when the maritime vehicle **100** is floating in a body of water.

[0092] The impellers **308** are disposed on the drive shafts **306**, respectively, and at least partly disposed in the transom channels **806**, respectively. As discussed above, each of the rotating impellers **308** typically includes several blades **552**. In some examples, the blades **552** are shaped (e.g., curved) to improve the hydrodynamic efficiency of the impeller **308**. As the impellers **308** spin, the impellers **308** impart momentum to the water, increasing its velocity.

[0093] The water, after passing through the respective impeller **308**, is directed into the respective diffuser **604**. The respective diffuser **604** gradually expands the flow area, converting the kinetic energy of the high-speed water into pressure energy. The pressurized water exiting the respective diffuser **604** is then directed through the specially shaped, respective nozzle **606** at the rear of the jet pump assembly **200**. As the pressurized water exits the respective nozzle **606**, the water creates a reactive force in the opposite direction, propelling the maritime vehicle **100** forward through the water. The respective nozzle **606** also includes the directional outlet **608** that can be tilted to control the direction of the water jet. By adjusting the angle of the directional outlets **608**, the operator can steer the maritime vehicle **100**.

Operating Efficiency of the Jet Pump Assembly

[0094] In some examples, the jet pump assembly **200** provides enough propulsive thrust for the maritime vehicle **100** to transition from a displacement mode or regime (e.g., the vehicle is mostly submerged and pushes water aside during movement) to a planing mode or regime (e.g., the vehicle lifts partially or mostly out of the water and may skim across the surface of the water). The hump speed or transition speed is when the maritime vehicle **100** transitions from the displacement mode to the planing mode. Below the hump speed, the vehicle **100** typically operates in the displacement mode, but above the hump speed, the vehicle **100** transitions into the planing mode. Conventionally, the efficiency of jet-based pump systems is proportional to the speed of the vehicle relative to the water, such that jet-based pump systems are typically less efficient in a displacement mode than in a planing mode. However, the jet pump assembly **200** described herein is maximally efficient in both a displacement and a planing regime.

Cooling System

[0095] FIG. 17 is a schematic diagram of an example cooling system **1700** that can be employed in the maritime vehicle **100** to reduce or maintain the temperature of various components of the jet pump assembly **200**, particularly the

motors **302** and the motor controllers for the motors (not shown). The cooling system **1700** generally includes a pump **1702**, a reservoir **1704**, and one or more heat exchangers **1706**. In some examples, the cooling system **1700** includes a single pump **1702** and a single reservoir **1704** fluidly coupled to one heat exchanger **1706**. In other examples, however, the cooling system **1700** can include one pump **1702** and one reservoir **1704** fluidly coupled to two or more heat exchangers **1706**, as illustrated in dashed lines in FIG. 17. Further, while the cooling system **1700** is described in connection with the jet pump assembly **200**, it will be appreciated that the cooling system **1700** can instead be used to cool the components of a different type of thrust system other than the jet pump assembly **200**.

[0096] In the present example, the pump **1702** pumps a coolant through an enclosed piping system **1708** that fluidly connects the pump **1702**, the reservoir **1704**, the heat exchanger(s) **1706**, and all other components of the cooling system **1700**. The piping system **1708** can include piping and various couplings and junctions (not shown) that enable the pump **1702** to pump coolant through the cooling system **1700**. For example, the enclosed piping system **1708** may include fittings to couple with couplings **422**, **424** of the cooling jacket **412** of each of the motors **302**, as described above. The enclosed piping system **1708** is preferably made of one or more materials that will not corrode or fail at the operating temperatures of the coolant and the cooling system **1700**. It will be appreciated that the coolant can include any known coolant. For example, the coolant may be a water and glycol solution.

[0097] In operation, the pump **1702** pumps the coolant from the reservoir **1704** through the jet pump assembly **200** so as to cool the electronic components of the jet pump assembly **200** as needed. More particularly, the pump **1702** pumps the coolant through, for example, the motor cooling jacket **412** of each of the motors **302** and/or a heat sink **1714** for each of the motor controllers (e.g., the electronic speed controller(s)). In some examples, the enclosed piping system **1708** can include controllable valves (not shown) to control how much coolant is distributed between the motor cooling jacket(s) **412** and the heat sink(s) **1714**. In some examples, the cooling system **1700** can be utilized to cool other components of the maritime vehicle **100** (e.g., a power source, other computer systems).

[0098] Heat generated by the motor(s) **302** and the motor controller(s) is transferred to the coolant passing through the motor cooling jacket(s) **412** and/or the heat sink(s) **1714**. As a result, the coolant is hotter after passing through the motor cooling jacket **412** and/or the heat sink(s) **1714**. The hot(ter) coolant then passes to the heat exchanger(s) **1706**. In some examples, the enclosed piping system **1708** includes a coupling **1722** before the heat exchanger(s) **1706** and a coupling **1724** after the heat exchanger(s) **1706**. The couplings **1722**, **1724** will be described in greater detail below.

[0099] Each heat exchanger **1706** includes an inlet **1728** that receives the hot(ter) coolant and an outlet **1732** at which chilled or cooled coolant is provided. In the present example, the heat exchanger **1706** can be a micro-keel cooler such as the micro-keel cooler described in greater detail in connection with FIGS. 18-22. In any event, during operation of the heat exchanger(s) **1706**, heat is transferred away from the hot(ter) coolant through the heat exchanger(s) **1706**. As described in greater detail below, the heat exchanger(s) **1706** convey(s) the heat transferred away from the

coolant into the water surrounding the maritime vehicle **100**.

[0100] In some examples, the couplings **1722**, **1724** are used to couple sensors to the cooling system **1700** to detect characteristics of the coolant. For example, one or both of the couplings **1722**, **1724** can include a pressure sensor and/or a temperature sensor. In some examples, the sensors send sensor signals to a controller that manages the operation of the pump **1702** to ensure the heat exchanger(s) **1706** operate(s) within acceptable ranges and, preferably, optimized temperature ranges.

[0101] Finally, the cooled coolant passes from the outlet **1732** of the heat exchanger(s) **1706** back to the reservoir **1704**. The reservoir **1704** is provided to accommodate volume expansion of coolant undergoing large temperature changes. The reservoir **1704** may also store extra coolant. In some examples, the reservoir **1704** can store an extra 33 percent (%) to 66% of the volume of the coolant flowing through the cooling system **1700**.

Micro-Keel Cooler

[0102] FIGS. 18-22 illustrate an example micro-keel cooler **1800** made in accordance with the present disclosure. In the present example, the micro-keel cooler **1800** is shown in use with the jet pump assembly **200**. In particular, the micro-keel cooler **1800** is disposed in a recess **1801** (shown in FIG. 15) adjacent to the jet pump assembly **200** installed on the maritime vehicle **100**, such that the micro-keel cooler **1800** is configured to cool the components of the jet pump assembly **200** when employed in the maritime vehicle **100**. In other examples, however, the micro-keel cooler **1800** can be used in connection with a different type of thrust system (e.g., a different jet-based pump thrust system) and/or a different maritime vehicle other than the maritime vehicle **100**.

[0103] With reference first to FIG. 18, the micro-keel cooler **1800** generally includes an inlet **1802**, an outlet **1804**, and means for reducing a temperature of fluid (e.g., coolant) flowing from the inlet **1802** to the outlet **1804**. In the present example, the inlet **1802** and the outlet **1804** are identical such that the micro-keel cooler **1800** is symmetrical. In other examples, the micro-keel cooler **1800** can be asymmetrical. Generally speaking, the means for reducing the temperature should be designed to maximize turbulent fluid flow. In the present example, the means takes the form of cylindrical tubes **1806** that are disposed between the inlet **1802** and the outlet **1804**. In other examples, however, the means can take the form of tubes, hoses, pipes, or other fluid conveying structures having a different cross-sectional shape (e.g., the means may include sharp internal corners of a triangle or a square instead of an internal circular shape).

[0104] As illustrated in FIG. 18, the inlet **1802** of the micro-keel cooler **1800** includes an inlet connection **1822** and an inlet mixing chamber **1824**. When the heat exchanger **1706** discussed above takes the form of the micro-keel cooler **1800**, the inlet connection **1822** receives the hot(ter) coolant after the coolant has passed through the motor cooling jacket **412** and/or the heat sink **1714**. The inlet mixing chamber **1824** is at least partially enclosed by an inlet endcap **1826**. The inlet mixing chamber **1824** ensures that the coolant received via the inlet connection **1822** is properly mixed before passing through the tubes **1806** to the outlet **1804**. The outlet **1804** includes an outlet connection **1832** and an outlet mixing chamber **1834**. When the heat

exchanger **1706** takes the form of the micro-keel cooler **1800**, the outlet connection **1832** is arranged to provide chilled or cooled coolant to the reservoir **1704**. In the present example, the outlet mixing chamber **1834** is identical to the inlet mixing chamber **1824**, such that the outlet mixing chamber **1834** is also at least partially enclosed by an outlet endcap **1826**. It will be appreciated that the tubes **1806** extending between the inlet **1802** and the outlet **1806**, the spacing of the tubes **1806**, and the total surface area of the micro-keel cooler **1800** facilitate the transfer of heat from the coolant as the coolant flows from the inlet mixing chamber **1824** to the outlet mixing chamber **1834** and out of the system via the outlet **1804**.

[0105] As shown in FIG. 19, the micro-keel cooler **1800** may further include an inlet bracket **1902**, an outlet bracket **1904**, and one or more plates **1906** coupled to each of the inlet bracket **1902** and the outlet bracket **1904**. The inlet bracket **1902** is arranged immediately adjacent the inlet **1802** (but on the opposite surface of the micro-keel cooler **1800** from the inlet **1802**). Likewise, the outlet bracket **1904** is arranged immediately adjacent the outlet **1804** (but on the opposite surface of the micro-keel cooler **1800** from the outlet **1804**). The plates **1906** extend outward (downward in FIG. 18B) from the plurality of tubes **1806** such that the plates **1906** are positioned to direct the water surrounding the maritime vehicle **100** over and between the tubes **1806**, which helps to further remove heat from the coolant flowing through the tubes **1806**. In the present example, the micro-keel cooler **1800** includes three plates **1906**, and each plate **1906** extends along the entire length of the micro-keel cooler **1800** between the inlet bracket **1902** and the outlet bracket **1904**. In other examples, however, the micro-keel cooler **1800** can include more or less plates **1906** and/or the plates **1906** can extend along only part of the length of the micro-keel cooler **1800**. In some examples, the plates **1906** can include or be coupled to one or more hydrodynamic surfaces that help to direct the surrounding water over one or more of the tubes **1806**. For example, the hydrodynamic surfaces may comprise an airfoil that directs the surrounding water between the tubes **1806**. The hydrodynamic surfaces can improve the efficiency of the micro-keel cooler **1800**. In some examples, the plates **1906** are simply snapped into the inlet and outlet brackets **1902**, **1904**. In other examples, the plates **1906** are fastened or welded to the inlet and outlet brackets **1902**, **1904**.

[0106] With reference now to FIGS. 19-21, the micro-keel cooler **1800** is assembled by coupling together a bottom plate **2000** including a first tube array **2002** (shown in FIG. 20), a top plate **2100** including a second tube array **2102** (shown in FIG. 21), and the inlet and outlet endcaps **1826** (shown in greater detail in FIG. 22). In the present example, the first tube array **2002** includes nine tubes **2002** and the second tube array **2102** includes eight tubes **2102**. In other examples, the first and second tube arrays **2002**, **2102** can include more or fewer tubes.

[0107] As best illustrated in FIG. 20, the bottom plate **2000** includes a first half **2012** of the inlet mixing chamber **1824**, a first half **2014** of the outlet mixing chamber **1834**, and the first tube array **2002**. As best illustrated in FIG. 21, the top plate **2100** includes a corresponding second half **2112** of the inlet mixing chamber **1824**, a corresponding second half **2114** of the outlet mixing chamber **1834**, and the second tube array **2102**. The first tube array **2002** is first welded or otherwise secured to the first halves **2012**, **2014**.

The second tube array **2102** is similarly welded or fastened to the corresponding halves **2112**, **2114**.

[0108] When the bottom plate **2000** and the top plate **2100** are coupled together, the halves **2012**, **2014**, **2112**, **2114** of the inlet and outlet mixing chambers **1824**, **1834** are coupled together. In the present example, the halves **2012**, **2112** and the halves **2014**, **2114** are coupled together via welding, adhesives, fasteners, sealants, or any combination thereof. Likewise, the first and second tube arrays **2002**, **2102** are coupled together to form the tubes **1806**.

[0109] The inlet mixing chamber **1824** and the outlet mixing chamber **1834** are capped, or enclosed, with the inlet and outlet endcaps **1826**, respectively. As shown in FIG. 22, each endcap **1826** includes a plug **2202** disposed on an end plate **2204**. The plug **2202** includes a sidewall **2206** that extends outward from the end plate **2204**. In the present example, the plug **2202** of each endcap **1826** is inserted into the respective mixing chamber **1824**, **1834** and the sidewall **2206** of that endcap **1826** sealingly engages an internal surface of the micro-keel cooler **1800** defining the inlet mixing chamber **1824** or the outlet mixing chamber **1834**. In some examples, the plug **2202** can further include welding or marine sealant to ensure the mixing chambers **1824**, **1834** are watertight.

[0110] The micro-keel cooler **1800** is assembled in the foregoing manner because the micro-keel cooler **1800** is too small to assemble in a more typical manner. For example, the tubes **1806** cannot be welded or fastened to the inlet and outlet mixing chambers **1824**, **1834** if the inlet and outlet mixing chambers are not manufactured in subsequently coupled halves as described herein.

Micro-Keel Cooler Cooling Capacity

[0111] In the present example, the maritime vehicle **100** includes two micro-keel coolers **1800**, as best illustrated in FIGS. 16 and 23. In the present example, the two micro-keel coolers **1800** are located on opposing sides of the keel **124**, with one of the micro-keel coolers **1800** immediately adjacent one of the sides **120** and the other of the micro-keel coolers **1800** immediately adjacent the other side **120** and separated by the jet pump assembly **200**. In this manner, one of the micro-keel coolers **1800** can cool the components of one of the jet pump drive systems **202** (more particularly the motor **302** of that jet pump assembly **202**), whereas the other micro-keel cooler **1800** can cool the components of the other jet pump assembly **202** (more particularly the motor **302** of that jet pump assembly **202**).

[0112] In some examples, the micro-keel coolers **1800** are capable of providing the total cooling capacity of the vehicle. In other examples, the micro-keel coolers **1800** supplement additional cooling systems (e.g., cooling exhaust systems) carried by the maritime vehicle **100**. In various examples, each micro-keel cooler **1800** is able to transmit approximately 1500 Watts to 2500 Watts of heat energy into the surrounding water. However, in some examples, each micro-keel cooler **1800** can improve heat transfer capacity and transfer as much as approximately 3500 Watts or 4500 Watts of heat energy into the surrounding water.

[0113] The micro-keel cooler **1800** is designed for improving heat dissipation while being small in size. In the present example, each of the tubes **1800** can have a diameter between approximately 0.25 inches (in.) diameter to 0.75 in. Additionally, each of the tubes **1800** can be between

approximately 6 in. and 36 in. in length. While the micro-keel cooler **1800** includes 17 tubes, in some examples, the micro-keel cooler **1800** can include more or fewer tubes. Moreover, while in the present example the micro-keel cooler **1800** includes two layers of tubes **1806**, in other examples, the micro-keel cooler **1800** can include one or more than two layers of tubes **1806**. Furthermore, while in the present example, the distance between the center of adjacent tubes **1806** is preferably between approximately 0.3 in. and approximately 1 in., it will be appreciated that the spacing between the tubes **1806** can be adjusted.

[0114] Additionally, the micro-keel cooler **1800** can further adjust the amount of heat dissipated into the surrounding water based on the coolant, coolant flow, and the conditions of the surrounding water. The coolant can be selected for an improved specific heat capacity. The coolant can, for example, have a specific heat capacity of between approximately 2 kilojoules per kilogram degree Kelvin (KJ/kgK) and approximately 5 KJ/kgK. Additionally, the coolant system **1700** is configured to pump between approximately 0.5 gallons per minute (gal/min) and approximately 3 gal/min of coolant through each micro-keel cooler **1800**. Further, the ambient environment of the vehicle can fluctuate between approximately -5 degrees Celsius ($^{\circ}$ C.) to approximately 40° C.

[0115] The dual jet pump assembly **200** and the cooling system **1700** incorporating the micro-keel cooler **1800**, as described herein, provide several benefits over other maritime vehicle power and cooling systems.

[0116] First, the dual jet pump assembly **200** can be operated with improved efficiency points. For example, the dual jet pump assembly **200** can operate efficiently in both a displacement regime and a planing regime. Furthermore, the dual jet pump assembly **200** can transition the vehicle from a displacement regime to a planing regime more efficiently than other systems. As a result, the dual jet pump assembly **200** causes the maritime vehicle **100** to be more versatile than other maritime vehicles.

[0117] Second, the dual jet pump assembly **200** is designed to be assembled as a lightweight, functional unit. For example, the dual jet pump assembly **200** can be manufactured and assembled prior to installation in a vehicle. This simplifies the installation process. Furthermore, the lightweight nature of the dual jet pump assembly **200** allows a single person to handle and install the dual jet pump assembly **200** in the maritime vehicle **100**.

[0118] Third, the dual jet pump assembly **200** can be installed as an operational unit. As a result, a technician installing the dual jet pump assembly **200** does not need to have access to an underside of the maritime vehicle **100**. The installation of the jet pump assembly **200** can be completed from an interior of the maritime vehicle **100**. Further, maintenance and replacement are simpler than other jet pump systems because maintenance can also be conducted from the internal cavity of the maritime vehicle **100**.

[0119] Fourth, the cooling system **1700** utilizes the micro-keel cooler **1800** to dissipate the heat generated by the dual jet pump assembly **200** into the environment (e.g., the water) surrounding the maritime vehicle **100**. The small size of the micro-keel cooler **1800** provides extensive cooling in a package or profile smaller than typical heat exchangers, which is advantageous given the small nature of the maritime vehicle **100**.

[0120] Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above-described examples without departing from the spirit and scope of the invention (s) disclosed herein, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept(s).

[0121] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

1. A micro-keel cooler for cooling a coolant flowing through a heat-generating component of a maritime vehicle, the micro-keel cooler comprising:

an inlet adapted to receive the coolant at a first temperature, the inlet including an inlet mixing chamber;

an outlet adapted to output the coolant at a second temperature less than the first temperature, the outlet including an outlet mixing chamber;

a plurality of tubes disposed between and fluidly coupled to the inlet mixing chamber and the outlet chamber and configured to direct the coolant from the inlet mixing chamber to the outlet mixing chamber; and

a plate coupled to the inlet and the outlet and disposed adjacent to the plurality of tubes, the plate configured to direct a fluid over at least a portion of the plurality of tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a cooling temperature that is less than the first temperature.

2. The micro-keel cooler of claim 1, wherein at least one of the inlet mixing chamber and the outlet mixing chamber is defined by a first half, a second half that is complementary to the first half and coupled to the second half, and an endcap coupled to the first half and the second half.

3. The micro-keel cooler of claim 2, wherein the plurality of tubes includes a first array of tubes and a second array of tubes, the first array of tubes welded to the first half and the second array of tubes welded to the second half.

4. The micro-keel cooler of claim 2, wherein the first half and the second half are welded together.

5. The micro-keel cooler of claim 1, further comprising a first bracket coupled to the inlet and a second bracket coupled to the outlet, wherein the plate is coupled at a first end to the first bracket and at a second end to the second bracket.

6. The micro-keel cooler of claim 5, wherein the plate is fastened at the first end to the first bracket and at the second end to the second bracket.

7. The micro-keel cooler of claim 1, further comprising a hydrodynamic surface coupled to the plate, wherein the hydrodynamic surface is configured to direct the fluid over the plurality of tubes.

8. The micro-keel cooler of claim 1, further comprising an inlet endcap coupled to the inlet to enclose the inlet mixing

chamber and an outlet endcap coupled to the outlet to enclose the outlet mixing chamber.

9. A method of manufacturing a micro-keel cooler for cooling a coolant flowing through a heat generating component of a maritime vehicle, the method comprising:

- providing an inlet and an outlet, the inlet adapted to receive a coolant at a first temperature and including an inlet mixing chamber, and the outlet adapted to output the coolant at a second temperature less than the first temperature and including an outlet mixing chamber;
- securing a plurality of tubes to each of the inlet and the outlet such that the plurality of tubes is disposed between and fluidly coupled to the inlet mixing chamber and the outlet mixing chamber, the plurality of tubes configured to direct the coolant from the inlet mixing chamber to the outlet mixing chamber;
- enclosing the inlet mixing chamber and the outlet mixing chamber with endcaps; and
- removably coupling a plate to the inlet and the outlet, such that the plate is configured to direct a fluid over at least a portion of the plurality of tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a cooling temperature that is less than the first temperature.

10. The method of claim 9, wherein providing the inlet and the outlet comprises coupling a first half of the inlet and/or the outlet to a complementary second half of the inlet and/or the outlet.

11. The method of claim 10, wherein the securing the plurality of the tubes to the inlet and the outlet is performed after coupling the first half of the inlet and/or the outlet to the second half of the inlet and/or the outlet.

12. The method of claim 9, wherein removably coupling the plate to the inlet and the outlet includes disposing a first end of the plate in a first bracket disposed on the inlet and disposing a second end of the plate in a second bracket disposed on the outlet.

13. The method of claim 9, further comprising coupling a hydrodynamic surface to the plate, wherein the hydrodynamic surface is configured to direct the fluid over the plurality of tubes.

14. A method of manufacturing a micro-keel cooler for cooling a coolant flowing through a heat generating component of a maritime vehicle, the method comprising:

- providing a first inlet half, a complementary second inlet half, a first outlet half, and a complementary second outlet half;
- coupling a first array of tubes to the first inlet half and the first outlet half;
- coupling a second array of tubes to the second inlet half and the second outlet half;
- forming an inlet by coupling the first inlet half to the second inlet half, the inlet adapted to receive a coolant at a first temperature and including an inlet mixing chamber;
- forming an outlet by coupling the first outlet half to the second outlet half, the outlet adapted to output the coolant at a second temperature less than the first temperature and including an outlet mixing chamber;
- enclosing the inlet mixing chamber and the outlet mixing chamber with inlet and outlet endcaps, respectively; and
- removably coupling a plate to a bottom of a plurality of tubes formed by the first array of tubes and the second

array of tubes, wherein the plate is configured to direct a fluid over at least a portion of the plurality of tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a cooling temperature that is less than the first temperature.

15. The method of claim 14, wherein forming the inlet comprises welding the first inlet half to the second inlet half and forming the outlet comprises welding the first outlet half to the second outlet half.

16. The method of claim 14, wherein removably coupling the plate comprises removably coupling a first end of the plate to the inlet and a second end of the plate to the outlet.

17. The method of claim 16, further comprising coupling a hydrodynamic surface to the plate, the hydrodynamic surface is configured to direct the fluid over the plurality of tubes.

18. The method of claim 16, wherein removably coupling the plate to the inlet and the outlet includes disposing the first end of the plate in a first bracket disposed on the inlet and disposing the second end of the plate in a second bracket disposed on the outlet.

19. A cooling system for cooling a heat-generating component of a maritime vehicle, the cooling system comprising:

- a reservoir adapted to store a coolant at a cooling temperature;
- a pump configured to pump the coolant from the reservoir through the heat-generating component of the maritime vehicle;
- a heat exchanger fluidly coupled to the heat-generating component and to the reservoir, the heat exchanger comprising a micro-keel cooler, the micro-keel cooler comprising:
 - an inlet configured to receive the coolant from the thrust assembly after the coolant flows through the thrust assembly such that the coolant has a first temperature greater than the cooling temperature, the inlet including an inlet mixing chamber;
 - an outlet configured to output the coolant at a second temperature less than the first temperature, the outlet including an outlet mixing chamber;
 - a plurality of tubes disposed between and fluidly coupled to the inlet mixing chamber and the outlet chamber and configured to direct the coolant from the inlet mixing chamber to the outlet mixing chamber; and
 - a plate coupled to the inlet and the outlet and disposed adjacent to the plurality of tubes, the plate configured to direct a fluid over at least a portion of the plurality of tubes to cool the coolant flowing from the inlet mixing chamber to the outlet mixing chamber, the fluid having a temperature that is less than the first temperature; and
- pipework configured to direct the coolant at the second temperature from the outlet of the micro-keel cooler to the reservoir.

20. The cooling system of claim 19, wherein at least one of the inlet mixing chamber and the outlet mixing chamber is defined by a first half, a second half that is complementary to the first half and coupled to the second half, and an endcap coupled to the first half and the second half, and wherein the

plurality of tubes includes a first array of tubes and a second array of tubes, the first array of tubes welded to the first half and the second array of tubes welded to the second half.

* * * * *